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MARIE SKŁODOWSKA-CURIE ACTIONS

Doctoral Networks (DN)
Call: HORIZON-MSCA-2024-DN-01-01

PART B

“BRAINET”

This proposal is to be evaluated as:
[DN]

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LIST OF PARTICIPATING ORGANISATIONS (max. 2 pages)

Consortium Member	Legal Entity Short Name	Academic (tick)	Non-academic (tick)	Awards Doctoral Degrees (tick)	Country	Dept./ Division / Laboratory	Scientist-in-Charge	Role of associated Partner or link to beneficiary
Beneficiaries								
Medical Center University of Freiburg	UKLFR	✓		✓	Germany	Department of Neurosurgery, Section Neuroelectronic Systems	Ulrich Hofmann	
Trinity College Dublin	TCD	✓		✓	Ireland	School of Engineering	Nicola Marchetti	
University of Essex	UESSEX	✓		✓	United Kingdom	School of Computer Science and Electronic Engineering	Michael Barros	
University of Freiburg	ALU	✓		✓	Germany	Institut für Mikrosystemtechnik	Matthias Kuhl	
South East Technological University	SETU	✓		✓	Ireland	Walton Institute for Information and Communications System Sciences	Indrakshi Dey	
Hamburg University of Technology	TUHH	✓		✓	Germany	Institute for Materials and X-ray Physics	Patrick Huber	
Technical University Ilmenau	TUIL	✓		✓	Germany	Institute of Biomedical Engineering and Informatics	Jens Haueisen	
Tampere University	TAU	✓		✓	Finland	Faculty of Medicine and Health Technology	Jari Hyttinen	
University of Minho	MINHO	✓		✓	Portugal	Center of Physics	Filipe Vaz	
Associated Partners								
NEUROCONN	NC		✓		Germany	EEG & tES divisions	Klaus Schellhorn	Hosting DC(s) on secondment
G.TEC MEDICAL ENGINEERING	GTEC		✓		Austria		Christoph Guger	Hosting DC(s) on secondment
Multi Channel Systems	MCS		✓		Germany	Research Projects and Predevelopment	Jannis Meents	Hosting DC(s) on secondment
MED-EL	MED-EL		✓		Austria		Markus Oberparleiter	Hosting DC(s) on secondment
Aalborg University	AAU	✓		✓	Denmark	Dept Health Science and Technology	Winnie Jensen	Hosting DC(s) on secondment
INSTITUT NATIONAL DE RECHERCHE EN INFORMATIQUE	INRIA	✓		✓	France	Models and Algorithms for Reliable Communication Systems	Malcolm Egan	Hosting DC(s) on secondment
PURPOSEFUL IDIOTIKI KE-FALAI OUXIKI ETAIREIA	PIKE		✓		Greece		Georgia Tsiliki	Hosting DC(s) on secondment
Delft University of Technology	TU Delft	✓		✓	The Netherlands	Faculty of Electrical Engineering, Mathematics and Computer Science	Dante Muratore	Hosting DC(s) on secondment

1 Excellence (starting on p.5) # @REL-EVA-RE@

1.1 Quality and pertinence of the project's research and innovation objectives

1.1.1 Introduction, objectives and overview of the research programme

1.1.1.1 Introduction

With an imminent greying of the European society more and more of our citizens will suffer from neurological disorders like Parkinson's Disease (PD), Dystonia, or Major Depression Disorder (MDD), currently treated in increasing numbers by Deep Brain Stimulation (DBS) through brain implanted stimulation electrodes¹. Although these implantations are done by key-hole surgery under stereotaxic control, they bear in addition to the usual surgical risks the acute risk of brain damage by haemorrhages and deep brain bleeding². Unfortunately, more concerning long term consequences have been observed like effects on human cognitive control and emotion–attention interaction potentially by off target, purely mechanical activation of neurons³. Adding to that, we recently showed in animal studies a connection between brain implants and oncogene activation, potentially causing Glioblastoma, the most devastating brain tumour⁴. Moreover, current state of the art DBS provides practically no on-demand control of the stimulation and does not take into account the state of the brain watering the stimulation's therapeutic effect down to some unnecessary brain intervention⁵.

In BRAINET, we confront the pressing need to keep clinical benefits of DBS whilst avoiding potentially devastating side effects of longest term brain implants.

As we tackle this formidable challenge, it is incumbent upon us to prepare the next generation of researchers, arming them with the knowledge, skills, and visionary insights necessary to drive novel technology to replace the present DBS into uncharted, but reliably safer and more flexible territories. BRAINET is an engine for excellence, convergence, and innovation. **BRAINET will lay the groundwork for doctoral candidates (DCs) to emerge as leaders in the scientific, technological, and industrial landscapes through establishing a training network for DCs and shaped by the evolution of distributed, and wireless deep brain interfaces.** To realise this vision, BRAINET has assembled a consortium, blending industrial pioneers with top-tier academic institutions. These partners collectively possess a rich tapestry of international expertise spanning the realms of DBS, neuroengineering, smart materials, networking, intra-body and wireless communications.

BRAINET sets out to unravel the complexities of **treating and managing neurological diseases** through the transformative concept of distributed wireless and interference brain stimulation. These efforts transcend conventional disease management paradigms, offering new dimensions of therapeutic potential. In the pursuit of temporal interference based non-invasive DBS, BRAINET embarks on the challenge of **harnessing the power of novel nanostructured smart materials in close contact with the brain.** These new materials hold the promise of efficacious brain stimulation and wireless energy harvesting⁶. The project is built as a global solution approach, starting from the stimulation electrode base-materials optimization by innovative technologies, including additive manufacturing, passing through smart surface functionalization protocols, where specially adaptive nanoarchitected thin films (conventional normal growth but also zigzag, spiral, inclined and combined growth geometries) and surface (coated and non-coated) plasma activation procedures will be optimised on the best-performing stimulation electrode base-materials to respond to conformal and flexible requirements of materials interactions with the brain⁷. The project will champion the exploration of **novel methods for temporal interference brain stimulation**, based on observed **brain semantics**. This journey takes us through in-vivo, in-vitro, and in-silico models, unveiling the power of brain control and interaction in previously uncharted territories. The creation of a semantic, **multiphysics wireless power and information transfer platform** is a keystone in ensuring the reliability, adaptability, and efficacy of wireless implantable devices. We propose a semantic communication-based model for the interplay between

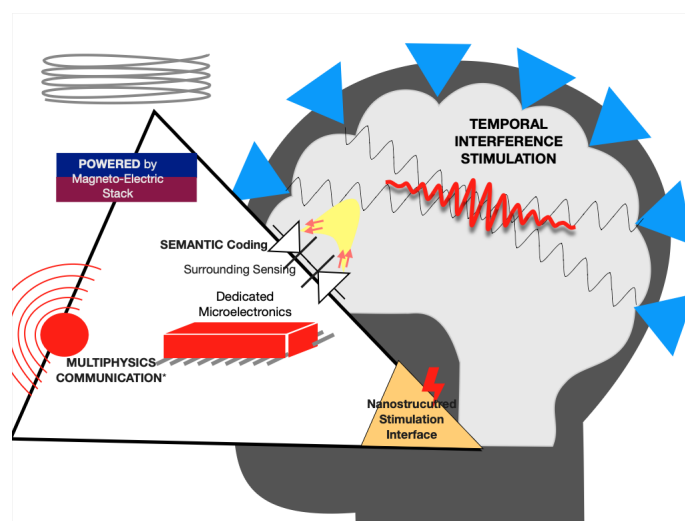


Figure 1: BRAINET Concept

of novel methods for temporal interference brain stimulation, based on observed brain semantics. This journey takes us through in-vivo, in-vitro, and in-silico models, unveiling the power of brain control and interaction in previously uncharted territories. The creation of a semantic, **multiphysics wireless power and information transfer platform** is a keystone in ensuring the reliability, adaptability, and efficacy of wireless implantable devices. We propose a semantic communication-based model for the interplay between

¹Visser-Vandewalle, V., et al. (2022), "Deep brain stimulation for obsessive-compulsive disorder: a crisis of access." *Nat Med.*

²A.J. Fenoy, R.K. Simpson Jr., "Risks of common complications ...," *Journal of Neurosurgery*, vol. 120, no. 1, pgs. 132-139

³Pfluger, P., et al. (2019). "Chronically Implanted Microelectrodes ..." *Front Neurosci* 13: 1367.

⁴Joseph, K., et al. (2021). "Transcriptional ... glial response ... flexible microprobes." *Biomaterials* 279: 121230.

⁵Tanskanen, J. M. A. et al. (2020). "Toward Closed-Loop ... A Review." *Bioelectricity* 2(4): 328-347.

⁶Nair, V. et al. (2023). "Miniature battery-free bioelectronics." *Science* 382(6671): eabn4732.

⁷Yue Shi et al., "Recent development... electrodes", *Smart Materials in Medicine* 1 (2020) 131–147.

wireless power and information transfer tasks for distributed interference stimulation, aiming to reduce the volume of data and energy consumption, increasing the number of supportable devices in the in-brain wireless network, and capture in a more meaningful and efficient way the goals of in-brain stimulation and recording. BRAINET is dedicated to **developing body-coupled networking ad-hoc protocols** that facilitate the coordination and efficiency of transcalvarial brain stimulators only touching the brain. We propose the novel principle of a decentralised, ad-hoc network of recording or stimulating micro-nodes, communicating like a sophisticated “Wi-Fi network” and taking advantage of distributed processing and communication. We introduce a network management framework for implantable solutions based on complex systems science and theory of autonomous self-organisation from computer networks. In such a framework, each node is a very simple autonomous agent with the ability to choose, e.g., modulation, black- or white-list neighbouring nodes based on local behaviour. Such autonomous network management can provide a reliable way to introduce self-organisation to the proposed wireless networking infrastructure, leading to the possibility of self-repair, when network nodes fail. The first of its kind BRAINET technology will be developed, tested and demonstrated with **in-silico simulations, novel in-vitro microphysiological systems and preclinical animal models of disease**.

BRAINET is equally **committed to nurturing a new generation of responsible researchers**. Our program encompasses comprehensive training for PhD students, including translational neuroscience, neuroethics, new smart nanoarchitected materials and wireless signal safety guidelines. We recognize that besides the technological development; we are pushing innovation that will reshape the landscape of bio-electronic medicine. BRAINET’s multidisciplinary team will lead us toward pioneering breakthroughs in communication engineering and neuroscience technology targeting clear unmet clinical needs and achieving substantial impact for better healthcare and societal benefits. *The BRAINET consortium includes top universities, research institutes, and industry partners. All partners are internationally renowned in DBS-related fields like neuroengineering, smart nanostructured materials, networking, and intra-body wireless communications.* BRAINET will chart a compelling path forward in the realm of neural interface technology.

1.1.1.2 Objectives

In addition to novel technology development and foreseen impact, **BRAINET vision is to ensure training tomorrow’s workforce to be ready to meet the urgent need for any type of brain stimulation as found in bioelectronic medicine and interventional psychiatry**. The progress of DBS is dependent on the design of novel wireless distributed implants, so the goal of the BRAINET Doctoral Network (DN) is to establish a training framework for doctoral candidates (DCs) who will contribute to the design and implementation of future most beneficial brain interfaces, laying the groundwork for them to become the future leaders in the resulting scientific, technological, and industrial fields. The DN’s mission includes developing a multi-disciplinary and cross-sectoral platform for research and educational training of “excellent” quality. The project will achieve the following ambitious and multidisciplinary research objectives (RO)s:

1. To research and disseminate information on the **treatment/management** of brain disorders and their emerging in-silico, in-vitro and pre-clinical neurological disease models by **distributed wireless stimulation** of the brain.
2. To research and develop new smart nanoarchitected materials for novel batteryless **transcalvarial implantable devices** that collect energy effectively and convert it into brain stimulation regimes.
3. To research and adopt novel methods for electric field temporal **interference brain stimulation enabling spatial targeting of deep brain structures based on observed brain semantics** demonstrated through in-vivo, in-vitro and in-silico models.
4. To research and develop a **semantic multiphysics power and information transfer platform for wireless implantable devices** supporting reliability, adaptability and interference stimulation.
5. To research and propose **body-coupled networking ad-hoc protocols** for Wireless Transcalvarial Networked Brain Stimulators (WTBS).
6. To create and support training for PhD students in translational neuroscience and medical device requirements, neuroethics and wireless signal safety guidelines for responsible, well-educated, and ethics-aware multipliers of knowledge.

1.1.1.3 Overview of the research programme

The overall DN programme comprises of seven WPs: **WP1** dedicates to project management, **WP6** to DC training and **WP7** to dissemination of knowledge. The four technical WPs will enable the achievement of the project’s scientific objectives and provide a tangible output by validating proof-of-concept use cases. **WP2** focuses on designing and developing the BRAINET technology, **WP3** is focused on how to develop

bidirectional communications when considering robust multiphysical communication network of implants and its management based on brain semantics, **WP4** will lay the groundwork for the semantical interference stimulation from materials to methods and finally **WP5** will provide in-vivo, in-vitro and in-silico test and validation of the technology developed. Furthermore, the WP structure is carefully designed to expose the DCs to all the different design skills and phases involved in applied and translational research. The approach and methodology adopted in each technical WP are explained in Sec. 1.2.

1.1.2 Pertinence and innovative aspects of the research programme

WP2 Wireless Transcalvarial Brain Stimulator (WTBS) Technology

State of the art (SOTA): Medical electrical implants became a significant force behind neuroscientific progress and developed from macroscopic pacemakers for clinical use – most notably Medtronic, Abbott-St. Jude, and Boston Scientific – to microscopic neural manipulators that range from simple yet dense passive micro-electrode arrays (Blackrock's bed of nails "Utah" arrays, Neuronexus' or Atlas' silicon multisite probes, Neuralink's polymer NeuraLace, Paradromic's microwire Connexus) to highly-integrated active neural interfaces⁸. Several research funding organisations outside Europe, especially in the USA, are investing in this technology e.g. DARPA or NIH-BRAIN initiative's SPARC funding. Yet, even the most promising efforts (e.g. Neuro Dust⁹, Neurograin¹⁰, or NeuraViPeR¹¹) **fail to deliver a clear roadmap to scale their solution to recording and stimulation channels towards future home care**. Even though some systems claim to be "fully immersible"¹⁰ with floating neural interfaces, a fully implantable approach that fulfils the longevity requirements of medical-grade implants is yet to be presented. Neuroscience researchers are also seeking ways to manipulate neurons without touching or penetrating tissue, including ultrasound or interference-based stimulation¹². Thus, recording and excitation of neural activity in the vicinity of a brain-machine-interface (BMI) have shifted to non-contact interference, as well as to the energetic and communicative domain. Energetic supply of implants has been demonstrated over nearly all physical near-field transmission channels¹³, among them electro-magnetic (inductive), capacitive¹⁴, optical, magnetic¹⁵, or ultrasound (US)¹⁶ transmission, or with direct local harvesting of (bio-)chemical processes, e.g. via bio fuel cells¹⁷. Data transmission resembles this list, with the necessity for either implant-side modulation of the carrier signal (inductive) or an independent radiofrequency (RF) generation to be able to establish bidirectional communication. Out of all these, only electro-magnetic (inductive or RF) realisations achieved clinical clearance, while the others suffer from reliability, complexity, or simply biological incompatibility. Most recently, as alternative to US and RF, magnetoelectric powering has been proposed for deeper (> 1 cm) deployment of stimulating devices, obviating tissue losses by low-frequency, purely magnetic drive fields, as well as body-coupled transmission¹⁸ utilising the body and its ionised substances themselves as a quasi-static transmission channel.

Beyond SOTA: The following innovations will be considered

Innovation 1: Building Wireless multiphysical communication ASICs (DC 1)

We explore inductive and magnetoelectric transducers for communication and innovative Body-Coupled Communication (BCC) for intra-body data transmission. We use X-Fab XH018 CMOS technology, which is ideal for medical implants, to ensure reliable design and fabrication. Our multiphysical, multiprotocol approach includes ionic and optical transmissions and demonstrates our dedication to developing a flexible, adaptable medical implant communication system.

Innovation 2: Bidirectional neural interfaces with arbitrary waveform stimulators (DC 2)

We will incorporate arbitrary waveform generators and analog-to-information neural recorders within our WTBS devices spread out in different areas that are used to make self-synchronized stimulation plans. We plan to programme sophisticated wireless addressing schemes and hand-shaking procedures between WTBS devices in hardware description language to create a distributed ad-hoc mesh with dynamically updated state machines imprinted into each WTBS. Synchronised neural stimulators with adaptable parameters to brain impedance variations are also being developed to improve stimulation accuracy and efficacy. LME's design library and X-Fab XH018 CMOS technology will be used to fabricate and test the ASICs, reducing risks and ensuring reliability.

Innovation 3: Development of subdural brain implants by additive manufacturing (DC 3)

This project pioneers the design and development of a flexible, conformal polymeric micro part for subdural

⁸Wendler, D. et al. (2021). "... scalable neural recording front-end for fully immersible neural probes ..." *IEEE ISSCC*

⁹Neely, R. M. et al. (2018), "Recent advances in neural dust...", *Current opinion in neurobiology*, 50, 64-71.

¹⁰Lee, J. et al. (2021). "Neural recording and stimulation using wireless networks ..." *Nature Electronics*, 4(8), 604-614.

¹¹Raducanu, B. et al., "128-channel neural stimulation and recording ASIC ...", *IEEE Eu-SSCC*, 2023.

¹²J. Cao et al., "Noninvasive Dynamic Patterns of Neurostimulation...", *IEEE Trans. on Biomed. Eng.*, 2020.

¹³Vishnu Nair et al., "Miniature battery-free bioelectronics." *Science*, 2023.

¹⁴Lee, C. et al. (2022). "A miniaturized wireless neural implant ..." *IEEE Journal of Solid-State Circuits*, 57(11), 3212-3227.

¹⁵Yuntao Lu et al., "Recent advances in power supply ...", *Journal of Micromechanics and Microengineering*, 2021

¹⁶M. Meng et al, "Design and optimization of ultrasonic wireless power transmission ...", *IEEE Trans. Biomed. Circuits Syst.*, 2017.

¹⁷K. Hayashi et al., "... CMOS supply modulated OOK transmitter ...", *IEEE Trans. Circuits Syst. II Exp. Briefs*, 2018.

¹⁸J. Lee et al., "Wireless Body-Area Network Transceiver ...", *IEEE Symp. VLSI Technology and Circuits*, 2023.

brain implantation using novel polymeric resins and manufacturing methods like Material Jetting and Vat Polymerization. The project advances the field by developing new, specialised polymeric resins with nanoparticles like gold and rigorously testing them for biocompatibility, mechanical strength, and long-term viability, including PVD nanostructured film coating. These implants are evaluated in vitro and in vivo to ensure their safety, reliability, and efficacy, marking a major advance in neural stimulation and recording technology.

WP3 Semantic and Multiphysics Communication for WTBS Networking

SOTA: With the advent of more flexible, distributed stimulation approaches (Neural Dust, -Motes, etc.) another bottleneck appeared: Slow backscattering communication with unique modes for each device is the main method of implementation that had to run along energy distribution channels. Even though communication is based on advanced signal processing techniques considering channel losses, interference, transmission and reception techniques, communication speed is always pitted against power transfer¹⁹. Several approaches have been studied as potential information carriers, but all contenders lack an overall advantage across several communication and powering metrics²⁰. Relying on single wireless media is likely to fail and hinder needed performance in pre-clinical settings¹⁸, even if e.g. Delta Encoding with its consecutive information interpretation might ease the bottleneck²¹. Without such and even more sophisticated feedback systems that adapt to the brain's semantics, this will likely lead to unpersonalized solutions^{22 23 24}.

Beyond SOTA: The following innovations will be considered

Innovation 4: Multiphysical communication to bypass head constraints (DC 4)

This project focuses on magnetic, NFC, and ionic communication mediums to revolutionise power and information transfer by overcoming the complex biophysical properties of a human head. We also want to look into magnetoelectric materials and how they might be used to get around electromagnetic shielding and tissue conductivity. This project aims to comprehensively characterise human heads' biophysical constraints to reveal the challenges. We then look to optimise multiphysical communication system using magnetoelectric, ultrasonic, and electromagnetic technologies. The project's multidisciplinary approach aims to create an efficient, safe, and practical communication system that will revolutionise networked power and information transfer for DBS and monitoring.

Innovation 5: Multimodal communication system for networked brain implants (DC 5)

This work will create a robust multimodal communication framework that integrates neural signals, electrical impulses, and other physiological indicators to connect brain implants and external devices. We will optimise data transmission and bandwidth using advanced signal processing and efficient encoding algorithms. Our method optimises system efficiency and practicality by reducing data transmission while preserving vital information. Finally, BRAINET tests biocompatibility and risks in vitro and in vivo to ensure safety, reliability, and long-term viability.

Innovation 6: Self-Optimizing and Self-Healing Implanted Communication (DC 6)

This project advances medical implant technology. Advanced Agent-based Modelling (ABM) models the implantable screw-like WTBS and their interconnecting links as individual agents in the networked system. This novel approach provides unprecedented insight and control over the network's properties, actions, goals, and behaviours. The project advances by self-optimizing computational power, memory, and energy resources in the near-field communication network to address complex resource allocation. To optimise network performance, data rate and energy availability are carefully monitored. Failure resilience and self-healing are the project's most innovative features. This project builds reliability and adaptability into implanted communication networks by predicting and responding to network failures.

Innovation 7: Semantics-based wireless power and information transfer (DC 7)

This groundbreaking project develops a semantic communication framework that intelligently samples and interprets brain functions and regions, transferring data and power based on significance and relevance. Beyond traditional methods, we use explicit and implicit communication strategies to transmit data and power only when needed and contextually relevant. For in-brain wireless network scaling, the project investigates novel centralised and distributed network architectures. This requires a sophisticated analysis of trade-offs between high-level decision-makers' information breadth and information and power transfer costs and delays. The project sets a new standard for wireless brain communication efficiency, effectiveness, and focusing on communication and power transfer semantics.

¹⁹Singer & Robinson. Advanced Healthcare Materials. 2021

²⁰Nair, V., et al. Miniature battery-free bioelectronics. Science, 2023

²¹D. De Dorigo et al., "Data Rate Reduction in Intracranial Digital Neural Probes using Delta Encoding", *IEEE BioCAS*, 2024.

²²Shi, G. et al. From semantic communication to semantic-aware networking... *IEEE Communications Magazine*, 2021

²³E. Uysal et al., "Semantic Communications in Networked Systems: ...," *IEEE Network*, 2022

²⁴P. Gorla et al., "Semantic-Functional Communications in Cyber-Physical Systems," *IEEE Network* 2024

Innovation 8: Coded information and sensing via backscattering control channels (DC 8)

This project uses coded information from an external transceiver to communicate with implants via a secondary wireless magnetic channel, improving stimulation configurations based on user-specified parameters. Compressed sensing will monitor implanted devices, providing a detailed per-device control channel for battery status and positioning for continuous, accurate monitoring. The multi-scale network-level control approach, which analyses power, time-delay, and coded stimulation commands with uplink monitored stimulation state information, makes the project stand out. This method helps understand implant interactions, particularly interference, bottlenecks, and information transfer delays.

WP4 Preparatory work for Semantics-based Spatio-Temporal Interference Stimulation

SOTA: Transcranial electrical stimulation (tES) features scalp fixed electrodes that are used to cause neurophysiological or behavioural correlates. Many tES protocols utilise a fixed experimental parameter set, disregarding individual anatomical and physiological properties. This one-size-fits-all approach might be one reason for the observed inter-individual response variability. Another reason is the considerable attenuation and blurring of the electric currents needed to pass through the skull in tES. While the attenuation and blurring will be overcome by the proposed BRAINET cross-skull (transcalvarial) applicators (implants), our patient-centric approach requires the simulation of current flowing through an accurate head model based on available individual anatomical (MRI, CT) data²⁵. Magnetolectric-based electronics then needs integration with novel distributed stimulation can be used brain-wide²⁶ through novel nanomaterials²⁷.

Beyond SOTA: The following innovations will be considered

Innovation 9: Current flow modelling in the head (DC 9)

This research will advance current flow modelling by using novel methods to dissect the skull, skin, and dura compartments of the human head in unprecedented detail. BRAINET combines CT and MRI with digital twin technology to improve brain current targeting. An uncertainty analysis measures the electrical conductivity of biological tissue, the uncertainty in compartmental geometry, and how these factors affect the electric fields created by implants. Finally, the project seeks optimal stimulation device placement. To achieve this, we will integrate the novel current flow models with the uncertainty analysis results using a novel optimisation approach that accounts for the implants' dynamic rearrangement potential, turning them into active agents in the search for optimal implantation sites.

Innovation 10: Control strategies of spatiotemporal Interference Stimulation (DC 10)

This research develops advanced neuronal system state measures and metrics to optimise **stIS**. The project will then test and validate these tools using an in-vitro 3D neuronal model platform to demonstrate its precision and accuracy in neural stimulation research. Integrating these cutting-edge assessment tools with spatiotemporal networked stimulation and communication technologies is the final goal. This integration of detailed neuronal system assessment and advanced stimulation techniques, shown in preclinical animal models, will revolutionise neural stimulation.

Innovation 11: Smart electrode materials for subdural brain implants (DC 11)

Smart coating architectures and surface plasma patterning will be optimized on avantgarde stimulation electrode base materials, aimed to respond to the particular requirements of the delicate and responsive interaction with the brain. An optimized material combination (base-material + coating) will be required to adapt to improved chemical, mechanical, and biological responses of electrodes by texturing them on a nanoscale level. Advanced methods like additive manufacturing will be used to and optimize the electrodes for conformality and flexibility to match the brain's contours. To increase brain reactivity and reduce biofilm formation and tissue injury, the project optimises surface properties like morphology and topography using complex bio-inspired shapes and smart manufacturing processes at the nanoscale level. The project also involves rigorous in vitro and in vivo safety, reliability, and long-term viability testing to minimise brain tissue risks.

Innovation 12: Hybrid magnetolectric energy harvesting (DC 12)

The project will develop a nanoporous silicon hybridised with polypyrrole (PPY) to convert small mechanical disturbances into electrical signals in biocompatible conditions and vice versa. This includes optimising porosity and functionalization, using hierarchically porous silicon for energy conversion. The second goal integrates this hybrid material with magnetostrictive elements like Metglas for efficient, wireless mechanical-to-electrical energy conversion. This method modulates magnetic field frequencies to deliver power and data to brain implants, a major advance in medical implant wireless communication. The final goal is to build and test MECMats, provide wireless energy to WTBS, and conduct in-vitro trials.

²⁵A. Paldanius et al., "Sensitivity Analysis ...Electrical Impedance Tomography ...," *IEEE Trans. on Bio. Eng.*, 2022.

²⁶Kim, W., et al. Magnetolectrics enables large power delivery to mm-sized wireless bioelectronics. *Journal of Applied Physics*, 2023.

²⁷Zhang, X., et al. Magnetic field driven catalysis of multiferroic magnetolectric nanocomposites. *Materials Lab* 2023

WP5 Integration and Demonstration of Networked WTBS (nWTBS) Technology

SOTA: BRAINET advances **stIS** by networking wireless neural interfaces towards a technology providing a significant step on the pathway to clinical-studies. This requires that many technical questions are explored, including (i) the basic mechanisms of the **stIS**' electric field affecting neuronal cells, (ii) the targetability of **stIS** in 3D brain-like environments (iii) how wireless neural interface requirements need to be defined, in terms of how to connect (network) WTBS nodes, their number, topology and functional flow to allow individualized stimulation strategies, and (iv) how recordings of neurons' electrophysiology can be performed under **stIS**. These recordings will then be used for guiding the stimulations to allow highest selectivity, resolution, and on demand based stimulation. We have already shown that 2D neuronal cultures can be used for **stIS** efficacy and control testing²⁸. **stIS** has also been demonstrated in preclinical²⁹ and in human scale models³⁰. There is also some emerging computer simulations on the **stIS**³¹. However, an in depth understanding of the effects of **stIS** on neuronal cells is still lacking. Advanced in-silico and in-vitro environments and their close interplay have the power to illuminate these and provide the understanding and design features for potential clinical applications and technologies.

Beyond SOTA: The following innovations will be considered

Innovation 13: : Self-organized 3D-in-vitro MEA for **stIS** (DC 14)

We develop complex 3D cultures, such as connected neuronal organoids or compartmentalised connected cultures with dedicated spatial areas for **stIS** stimulation including areas of highly interconnected neuronal cells and areas of directional axons. These will be replicated in-silico to transfer knowledge to in-vivo models and further to human scale. The developed WTBS nodes with both stimulation and recording properties will be tested in the developed 3D in-vitro setups, in preclinical animal models and in-silico simulations in human scale. Optimising these networked node electrode arrays to characterise stimulation's spatial and temporal properties and neuronal responses will provide the validation of the technology and later help design human in-vivo systems and multimodal communication technologies. This project comes to an end with the integration and demonstration of a complex multimodal communication system and a self-organized sensory network in a 3D microphysiological environment.

Innovation 14: Parkinsonian Rat Model Pre-clinical Validation of **stIS** (DC 13)

This study starts with unilateral stereotaxic injections of 6-OHDA to make a strong hemi-Parkinsonian rat model. This model allows detailed mobility pattern analysis, laying the groundwork for treatment efficacy evaluation. Deep electrodes are used to measure intracortical electrophysiological changes using superficial **stIS**, a novel way to study xneurostimulation mechanisms. The project compares DBS and **stIS** using advanced electrical stimulators for multisite stimulation. It also uses advanced histological and -omics analyses to study **stIS**'s physiological and genetic effects at the cellular and molecular levels. Wireless neurostimulation technology advances by validating wireless WTBS in hemi-Parkinsonian rats in the final stage.

1.2 Soundness of the proposed methodology

The research methodology of BRAINET is divided into three phases:

Phase 1 – Requirement Gathering and Design: This initial phase provides the necessary definitions and critical decisions on the specific technical approaches concluded from pre-clinical insights. It includes the description of requirements, use cases, architecture and roadmaps for the required technological developments.

Phase 2 – Research and Development: This phase forms the core of the project activities and relates to developing the technology proposed in the project. The relevant WP2-5 focus on detailed experimental design, material selection and optimization, development of prototypes or models, and initial testing under controlled conditions. This phase is critical for turning theoretical concepts into practical applications, involving iterative processes of design, testing, and refinement.

Phase 3 – Proof of Concept and Evaluation: The final phase is dedicated to validating and demonstrating the effectiveness of the BRAINET technology. This involves comprehensive testing in in-vivo, in-vitro and in-silico, and evaluation of the system's feasibility and potential for broader application. Feedback from this phase is used to make final adjustments and improvements, culminating in a proof of concept that showcases the project's success and lays the groundwork for future development and implementation.

1.2.1 Overall Methodology

WP2 (Lead - ALU): Wireless Transcalvarial Brain Stimulator (WTBS) Technology

²⁸Ahtiainen, A. et al., (2023) "Stimulation of Neurons and Astrocytes via Temporally Interfering Electric Fields." *bioRxiv*, 2023-10.

²⁹Wessel, M. J. et al. (2023). "Noninvasive theta-burst stimulation ... motor skill learning." *Nat Neurosci* 26(11): 2005-2016.

³⁰Violante, I. R. et al. (2023). "Non-invasive temporal interference electrical ... of the human hippocampus". *Nature neuroscience*.

³¹Karimi, F. et al. (2019). "...deep brain stimulation based on interfering electric fields". *Physics in Medicine & Biology*.

Aims: Our objectives encompass the design of a centralised command unit, an efficient communications stack, a robust environmental sensing unit, a physical communication layer, mechanisms for internal and external energy and data transmission, and an advanced module for tissue coupling. **Challenges:** Advancing wireless brain stimulator technology presents multifaceted challenges, especially for distributed implants. A primary obstacle is pinpointing biophysically pertinent parameters, considering the brain's intricate inhomogeneous mechanical and electrical properties. Material fabrication poses its own set of challenges: efficiently translating magnetic energy from outside the brain tissue to electrical energy within, all through wireless means, requires precision. Furthermore, the synthesis of a bidirectional neural interface in CMOS technology, potentially complemented by a neural-data pre-processor, embodies its own set of intricate challenges.

Methodology: Partners will collaborate to identify biophysical parameters and biological considerations in BRAINET methodology. We start by creating a cutting-edge WTBS network communication architecture. We will wirelessly convert magnetic energy to electrical energy since the implants are batteryless and need to harvest energy. We use PPy-Si with magnetostrictive materials like Metglas to exploit silicon's piezoresistance. During communication hardware development, we design a logical transmission protocol for magnetic, NFC, and ionic communication with optical feedback (DC1). We'll build and demonstrate WTBS with CMOS circuitry. We are pioneering intra-body bidirectional data transmission for external communication using inductively linked phase and amplitude modulation (DC1,2), Arbitrary waveform generators and timing references are tested on FPGAs before they are built into an ASIC. Synchronised neural stimulators adapt to brain impedance changes (DC2). WTBS devices will use additive manufacturing to develop subdural brain implant polymeric materials. Testing for chemical, mechanical, and bio-functional properties ensures brain implant materials meet strict standards (DC3).

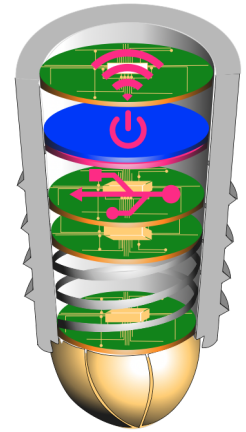


Figure 2: A single WTBS

WP3 (Lead - UESSEX): Semantical and Multiphysical Communication for WTBS Networking

Aims: We aim to conceive, test, and verify a communication system that seamlessly integrates the external interface with the WTBS. Concurrently, we will integrate and manage of multiple implants within a singular network. We will improve communication by using multiphysics, multimodal, and semantic communication theory, coding, and management. We aim to achieve a semantic communication theory-based framework, targeting coded information transmission based on stimulation configuration and compressed sensing to achieve implanted devices' status monitoring.

Challenges: Paramount among these is navigating the biophysical constraints and requirements posed ultimately by the human head and body, particularly during the intricate processes of intra-body and body-coupled communications. An overarching concern is the need to balance reliable communication with energy conservation, a critical aspect for implantable devices. As we venture into the realm of expanding the network, data collision, where multiple devices communicate simultaneously, emerges as a potential hurdle. We need to ensure that the proposed in-brain multimodal wireless communication system achieves satisfactory performance, stability, and safety. Magnetoelectric materials can help overcome biophysical human head limitations, including the skull's electromagnetic shielding effect, tissue conductivity, and electromagnetic absorption rates. We wish to minimise the volume of data and energy consumption while securing scalability, real-time monitoring, and stimulation performance. Synchronising the communication operation of a potentially large-scale network of WTBS we have to consider self-organisation properties. We need to address the ability to overcome node failure, either physical or logical.

Methodology: Never-before-seen intra-body and body-coupled communication systems will be newly designed, modelled, implemented and evaluated in multiple use cases. Magnetoelectric, NFC, and ionic communications with optical feedback will be tested. This will involve a thorough study of magnetoelectric materials and their potential to overcome the skull's electromagnetic shielding effect. We use magnetoelectric communication to model human heads' biophysical limitations and their effects on power and information transfer. The multiphysical system will be optimised for networked power and information transfer, taking into account human head biophysical limitations, using feedback loops (DC4). We will create a reliable and efficient WTBS multimodal communication system (DC5). Signal processing, data compression, and efficient encoding algorithms will optimise multimodal communication system data transmission to maximise bandwidth and minimise power consumption. Safety, reliability, and long-term viability of the WTBS multimodal communication system depend on this. We will create a semantics-based framework to sample brain functions and regions and transfer data and power. We will examine both explicit and implicit communication, including information related to brain inactivity (DC7). Early on, we will use computer simulations and models to design intra-body and body-coupled communication systems and make initial adjustments and refinements.

The BRAINET approach will enable multi-scale network-level control, allowing us to monitor how different implants affect each other, such as interference, information transfer bottlenecks, and delays (DC8). We will combine CMOS technology with multi-physical transducers to create a robust communication architecture. We will conduct rigorous electrical tests, focusing on data collision scenarios, to ensure the robustness and reliability of our final product (DC2). Based on user-specified parameters, the external transceiver transmits coded stimulation configuration information to the implants over a secondary wireless magnetic channel (DC8). We will use complex systems science, particularly ABM, to model a network of WTBS nodes with computational agents for screws and links. Agents will manage resources (e.g., computational power, memory, power) to optimise network performance indicators like data and information rate, energy storage, and availability in the in-brain wireless network (DC6). WTBS will be connected through a small-scale near-field communication network. Failure resilience and implanted network self-healing will be used (DC6).

WP4 (Lead - TUIL): Preparatory Work for Semantics-based Interference Stimulation

Aims: The overarching intent of our research initiative, titled "Semantics-based Interference Stimulation," is to pioneer and standardise semantics-based electrical interference stimulation through a network of cranial implants. Our five-fold objectives are: (a) To revolutionise the current paradigms of flow modelling by innovatively modelling the skull, skin, and dura compartments. (b) To extract optimal parameters for stimulation by utilising these new models and conducting an uncertainty analysis concerning the electrical conductivity and compartmental geometry of the biological tissues. (c) To pinpoint optimal implantation sites within the rodent model of Parkinson's Disease. (d) To institute and authenticate tethered interference stimulation within the rodent PD model. (e) To authenticate our theories on semantics-based interference stimulation via proof-of-principle experiments.

Challenges: While our project is ambitious, it is also fraught with complex challenges that demand nuanced solutions. First, the process of integrating high-resolution imaging with digital twins to build sophisticated head models is inherently intricate. Uncertainty analysis, given the unpredictable nature of biological systems, poses a significant challenge in ensuring the reliability and effectiveness of the implants. Positioning the implants optimally requires a delicate balance between maximising effectiveness and minimising invasiveness. Further complications arise from ethical considerations, especially when establishing and working with animal models of a disease as profound as Parkinson's. We also have the monumental task of ensuring that our proof-of-principle experiments are not just successful but also indicative of potential success in larger, real-world applications. **Methodology:** Interference stimulation begins with current flow modelling for the desired brain implant technology. This requires novel methods to model skull, skin, and dura compartments with more subcompartments. We aim to target brain currents optimally using CT and MRI and implant digital twins. An uncertainty analysis will quantify the impact of biological tissue electrical conductivity and compartmental geometry on the implanted electric field. Finding optimal implantation sites is another important part of our research. This approach considers rearranging implants as dynamic agents to find optimal implantation sites (DC9). We develop neuronal system state metrics. This is essential for interference stimulation to maximise benefits and minimise side effects. We use signal-based, spatial, and brain-dynamic state assessments to evaluate stimulation needs and outcomes. We evaluate and demonstrate these tools using an in-vitro 3D neuronal model platform with spatiotemporal stimulation. (DC10). Complex bio-inspired electrodes (base material) and rapid manufacturing are used. A major focus of our research is on smart coating nanoarchitectures and surface plasma patterning of these bio-inspired electrodes for WTBS aiming to improve the implants' chemical, mechanical, tribological, bio-function, and anti-infection responses. Innovative coating and plasma-based surface treatment methods will focus on nanoscale surface texturing and systematic surface optimisation to achieve safety, reliability, and long-term performance of functionalized electrodes. (DC11). We also are developing nanoporous silicon hybrid materials with magneto-to-electrical energy materials for efficient energy conversion. This allows wireless energy and data transmission to brain implants (DC12).

WP5 (Lead - TAU): Integration and Demonstration of Wireless Transcalvarial Brain Stimulators

Aims: Our core endeavour within this workpackage is to curate the first-ever multiscale and multimodal demonstration environment for Networked WTBS systems. We aim to: (a) Craft in-silico, in-vitro, and pre-clinical environments for development, testing, and showcasing individual components, and the entire system to seamlessly integrating them into a larger networked brain simulation architecture. (b) Utilise the multifaceted environments for the development and evaluation of our project concepts, steering our technologies towards an innovative integrated solution. (c) Offer a holistic demonstration of our meticulously developed technology components, encompassing interference stimulation, schematics-based stimulation, multimodal communication, and state-of-the-art power and data transfer technologies.

Challenges: Establishing first-in-the-world technologies brings forth a multitude of challenges. Foremost is the need to produce precise computer simulations that accurately mimic a range of environments

to develop the electric field interference stimulation and the novel data and power transfer and communication technology—from in-vitro to human scale. The design of a novel 3D spatially distributed in-vitro neuronal system where the spatio-temporal interference stimulation can be tested on neuronal cells and their parts such as somas and axons, requires deep understanding of the in-vitro methods on neuronal engineering to ensure validity and reliability. In tandem, optimising the simulation-based in-vitro and in-vivo environments and the interrogation technology to assess the resulting neuronal effects demands a synergy of our advanced tech and biological insights. Our challenging ambition is to integrate the novel components, each with its own intricacies, into a cohesive system in-vitro and in-vivo. We will also tackle the ethical considerations of the biological setups ensuring the welfare of animals during in-vivo testing.

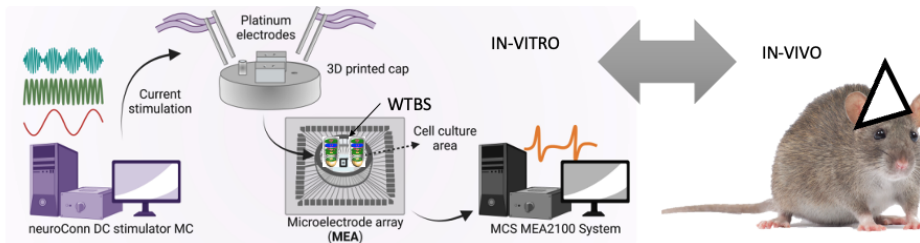


Figure 3: WP5 methodology validating stIS in-vitro, in-silico, and in-vivo models

light sheet microscopy to test the technology, measure the neuronal topology and function. This lets us test neuronal responses thoroughly, find out about their activity, and demonstrate how the system works. We optimise electrode placements and data and power transfer interfaces in-silico, in-vitro, and in-vivo. (DC14). In-vivo testing of innovative stimulators in our approved rodent model for Parkinson's Disease based on stereotaxic 6-hydroxydopamin lesioning will evaluate their therapeutic efficacy. We will assess stIS-induced intracortical electrophysiological changes. We will measure the driving fields, local field potentials, and action potentials produced by stIS with superficial and depth electrodes. Quantitative analysis of animal behaviour will determine the efficacy of these treatments. We also want to test early WTBS nodes in PD animals. We will use alternating external magnetic fields to power wireless WTBS on hemi-Parkinsonian rats. The effectiveness of these treatments for PD symptoms will be evaluated through movement challenges and histological analyses to determine long-term effects (DC13).

1.2.2 Integration of methods and disciplines to pursue the objectives

The research program fosters cross-domain collaboration across WPs 2-5, with all DCs linked to WP5 for theoretical and practical integration. It provides complementary research and training, blending academic and industry expertise through a diverse, interdisciplinary network, and facilitates secondments for industrial exposure. Renowned academic research groups from all across Europe active in the domains of the design and development of communication systems, UESSEX, TCD, SETU and ALU, will synergise to train and supervise the DCs participating in BRAINET. UKFLR, TAU, TUIL have all competence in the neuroengineering domain, both in-vitro and in-vivo. TAU and TUIL have an established track record on the computational side with regards to neuro- and interference stimulation. MINHO and TUHH have a proven record in biomaterials for implantables, including energy harvesting for various functionalities. All these partners will be providing training in their respective fields of expertise. DCs will also benefit from inter-sectoral research and training experience from five industry associated partners (NEUROCONN, G.Tec, Multi Channel Systems, MED-EL AND PURPOSEFUL). Other associated partners (AP) with complementary skills include Aalborg University and its experience in pre-clinical implantable models, INRIA and its experience in modelling complex systems for information and power transfer as well as Delft University of Technology and its experience with hardware for intra-body communications. The APs have extensive know-how and world class research infrastructures (see Section 3.2.1 and Section 7).

1.2.3 Gender dimensions and other diversity aspects

BRAINET favours the inclusion and access to stakeholder voices in the design process, and this cannot be achieved without including all genders. Guidance with reference to gender questions will be informed by the EU 'Toolkit Gender in EU Funded Research'.³² The **Stanford University 'Gendered Innovations'** resources will also be used throughout: These resources include specific steps for incorporating sex and gender analysis in co-design. The following measures will be taken in this research:³³ Considering the gender dimension and diversity aspects in the research's impact on quality of life, BRAINET will adopt the following policies: **Gender Impact Assessment (GIA)** - The GIA outlined by the European Institute for Gender Equality³⁴ will be conducted. A five-step assessment using gender and intersectional analysis evaluates potential impacts

³²<https://op.europa.eu/en/publication-detail/-/publication/c17a4eba-49ab-40f1-bb7b-bb6faaf8dec8>

³³<https://genderedinnovations.stanford.edu/>

³⁴<https://eige.europa.eu/>

during research and before final decisions. **Communication and Training** - All documents and outputs will reflect equity and gender-neutral language, with a gender-sensitive approach applied to research, writing, and dissemination. Team members will undergo gender equity training when available. **Gender Diversity** - BRAINET will consider the gender differences that could be associated with the proposed project such as differences in head shape and brain structure, composition and distribution of neuronal cells etc. For example, the corpus callosum, the bundle of nerve fibers connecting the two hemispheres of the brain, has been found to be proportionally larger in women. This could potentially influence the spread of electric fields in the brain and thus needs to be included in human scale stIS simulations and WTBS deployment. The distribution of neurotransmitters and receptors, which play a crucial role in neural communication, differs between males and females, influencing cognitive and emotional processes. Research has suggested that men and women may rely on different brain networks to perform certain cognitive tasks, potentially leading to variations in cognitive strategies and outcomes. Hormonal differences between genders, particularly the effects of sex hormones like estrogen and testosterone, also impacts brain function and structure. Understanding gender nuances is vital for inclusive projects like BRAINET, ensuring interventions and analyses account for gender-specific factors affecting outcomes.

1.2.4 Open science practices

In accordance with the European Commission policies, BRAINET supports the Open Access publishing mandated by Horizon Europe. Other forms of open access dissemination will be heavily utilised by BRAINET, such as (a) the release of pre-publication versions via the website. (b) Conference paper presentations are made available online. (c) Sharing online resources, such as papers, models and slideshows, via federated social media. The consortium plans for both "green" (self-archiving in one or more repositories) and "gold" (paying an authors' fee when publishing in journals) forms of publishing. All major international publishers (including IEEE and OSA) provide several models for Open Access publishing, with various options on what licence to place on the published documents.

1.2.5 Research data management and management of other research outputs

The BRAINET project consortium is fully aware of the European legislative context (EC directives 2002/58/EC and 95/46/EC and EU 2016/679 GDPR). In terms of privacy and data security, the project does not raise any concerns. All of the data used in the project will come from machines in the environment and will not include any information that could be used to contact or locate a specific individual. Gathering sample data for training and knowledge transfer activities (such as the number of website visits, the average download time for deliverables, the number of leaflets distributed, the number of participants, the feedback received, etc.) can improve dissemination efforts. BRAINET will also develop a code of practice from the start of the project to specify the type of data used and stored, informed consent procedures, and secure data storage and transfer, which will be agreed, signed, strictly implemented, and reviewed annually at PMB meetings to ensure EC directive compliance.

1.3 Quality and credibility of the training programme

1.3.1 Overview and content structure of the doctoral training programme

BRAINET DN will hire 14 DCs within the maximum period of the project 48 months with each DC's contract being 36 months. The DCs will enjoy during their appointment period up to 6-month secondment phases at the premises of one of the associate partners in order to foster and strengthen collaboration among partners. The secondment will be mostly cross-sectorial. That is, except in few cases, DCs originally hosted by academic institutions will be seconded to nonacademic partners. DCs will thus be exposed to both industrial and academic research environments, broadening their research vision and positioning them to advance in their careers. The appointed DCs will be overseen by a local supervisor(s) at the hosting partner, a supervisor at the partner that plans to second the DC, and the consortium Supervisory Board (SB). Part B2 defines the specifics of the SB. The DCs must be actively involved in the hosting institution's research activities. The main Supervisor(s) will devote at least 15% of their time to overseeing the recruited DCs on a daily basis, while the SB's responsibility will be to maintain a holistic overview to ensure that acceptable levels of performance are achieved by all DCs across the Network. Personal Career Development Plan (PCDP): Each DC will define a PCDP with their Supervisors to shape their training programme and determine goals to be met. The PCDP must be completed and approved by the SB within the first two months of the DC's contract. The PCDP will define the main research topics and will include the names and roles of the Supervisors. A list of required courses and training will be identified based on the background of each DC. The PCDP will also clearly state a timeline for the progress of the DC research, as well as milestones and evaluation indicators,

allowing the SB to track the progress of individual DCs. To evaluate the performance of the DCs, quality and quantity indicators will be defined and included in the PCDP. Individual DCs' mandatory courses and the number of required published papers in journals and conferences, for example, will be defined. The PCDP will include a publication plan that will define targeted journals and conferences to ensure a certain level of academic outreach. Other indicators, as long as they are approved by the SB, can be defined and used. As part of the training, the DCs must follow one existing research project, which will be named in the PCDP. The DCs' secondment plan will be detailed in the PCDP. The plan will include a visit to the institution, the date of secondment, the period, and the contact at the visiting institution. A provisional secondment plan is already provided in Section 3.1. The plan will pinpoint the needs and goals of training and how to achieve them. The PCDP will be revised annually, and modifications may be applied, if needed. PCDP will also include the training events as in Table 1 and related Academic Programs in what follows.

Table 1 Main Network-Wide Training Events, Conferences and Contribution of Beneficiaries

	Main Training Events & Conferences	ECTS	Lead Institution	M
1	Workshop on Neural Engineering and bioelectronics		TAU	16
2	BRAINNET Conference		UKLFR	30
3	Showcase event in a major conference (IEEE)		TUIL	42
4	Transectorial Dissemination day event 1		UESSEX	20
5	Transectorial Dissemination day event 2		SETU	24
6	Transectorial Dissemination day event 3		TCD	46
7	Team building event		MINHO	6
8	Course 1: Writing Scientific papers and presentation skills	if possible	UESSEX	8
9	Course 2: Commercialization of biomedical devices and software	if possible	TAU	10
10	Course 3: Preparing for pre- and clinical trials	if possible	UKLFR	12
11	Course 4: Ethics in a practical way	if possible	UKLFR	24
12	Course 5: Exploring novel tools in grant writing applications	if possible	UKLFR	32
13	Course 6: Team building experience	if possible	ALU	40
14	Course 7: Research Integrity and Impact in an Open Scholarship Era	if possible	TCD	14
15	Course 8: Planning and Managing your Research and your Career	if possible	TCD	26
16	Course 9: Getting value out of a PhD beyond writing papers	if possible	SETU	34
17	Course 10: Working with others: team-work and inclusion	if possible	SETU	22
18	Course 11: Hygiene and safety in laboratory activities	if possible	MINHO	10
19	Course 12: Monitoring and management of large projects	if possible	MINHO	18
20	Course 13: Research data management	if possible	TUIL	11
21	Course 14: Biocompatibility and biosafety of implants	if possible	TUHH	20
22	Seminar 1: Principles of wireless neural interfaces		UESSEX	7
23	Seminar 2: Bioelectronics concepts for neuronal interfaces		TAU	9
24	Seminar 3: The foreign body response from brain implants		UKLFR	11
25	Seminar 4: Information Processing for Cyber-physical Systems		SETU	21
26	Seminar 5: Complex Systems for Intra-body Networks		TCD	15
27	Seminar 6: Alternative Computational Methods for Future Networks		TCD	17
28	Seminar 7: Introduction to CMOS circuit layout and its impact		ALU	19
29	Seminar 8: Multidisciplinary Optimization Methods		SETU	21
30	Seminar 9: Nanoarchitected Thin Films: applications to materials surface functionalization		MINHO	27
31	Seminar 10: Temporal Interference Stimulation		TUIL	13
32	Seminar 11: Multifunctional Materials		TUHH	33
33	Summer School 1: Electrophysiologic recordings from living sources	if possible	UKLFR + ALU + TUHH	20
34	Summer School 2: Introduction to Wireless Intrabody Comms	if possible	TCD + SETU + UESSEX	13
35	Summer School 3: Electric Interference Stimulation	if possible	TAU + TUIL	33
36	Winter School: Additive Manufacturing: from lab to factory floor	if possible	MINHO	24
37	Startup simulation roleplay	if possible	UKLFR	24

The proposed Training program shall constitute three major elements (Basic Training, Research Activities and Extra-curricular Training): **Basic Training:** PhD level courses, tutorials, seminars, workshops and conferences. The offered material must be closely related to BRAINNET's main research track. **Research Activities:** DCs will be required to conduct one specific research project within the host institutions. The topic of the research project must fit within the Network's defined research areas. However, the DCs will have the opportunity to follow multiple ongoing projects within the Network that are relevant to their research topics. DCs will have the opportunity to be exposed to various research environments as well as participate in hands-on research activities as part of this project. **Extra-curricular Training:** The training framework will also include additional courses that will provide DCs with valuable skills for future promising careers. Leadership skills, presentation skills, writing scientific papers, patent filing, IPR issues, business planning, and proposal writing are examples of complementary topics/skills. Local courses (e.g., post-graduate courses, seminars), research projects, and expertise are available to DCs at their hosting institution. Secondments and visits, on a broader level, provide another opportunity for DCs to benefit from different expertise, courses, and seminars. Finally, specific partners will organise network-wide training activities (plenary meetings, seminars, workshops, summer/winter schools, etc.) and make them available to all DCs in the Network. As a result, the entire training programme is designed with the following multi-dimensional goals in mind: i) provide DCs with essential and cutting-edge research state-of-the-art knowledge (via seminars, workshops, integration within

organised research groups, etc.); ii) foster DCs' skills and practical knowledge to increase the impact of their research work and future employability (via extracurricular activities, etc.); and iii) establish the foundation for each DC to create their future professional network (through plenary meetings, conferences and workshops, secondment, etc.).

The components of the planned doctoral training will cover the following activities (other activities can be proposed and adapted if considered useful by the Network expertise):

1. **Research:** DCs will take part in various research activities as defined by BRAINET's technical work packages. Each DC will be assigned one or two research topics to contribute to, with the local Supervisor(s) providing close supervision and collaboration. Traditional academic procedures will be used in the research activities, such as surveying the state-of-the-art, defining drawbacks and challenges, acquiring necessary scientific knowledge, proposing new methods, and so on. Practical implementation of proposed methods at participating partners will also be part of the research activities. Each DC, in collaboration with their Supervisor(s), will determine the research field and related topics to be studied, which must be approved by the SB to ensure alignment with the network's research topics and the diversity of topics covered by the Network. The SB will be in charge of ensuring the quality of results and achievements. The criteria for selecting Network partners ensure that the consortium partners have the necessary background to carry out the proposed research activities within the Network and will provide the necessary training activities for the DCs.

2. **Academic Programs:** DCs will benefit from existing local training capabilities as well as a global view of the entire network. As beneficiaries of the Network, nine academic partners (UKFLR, ALU, UESSEX, TCD, SETU, TUIL, TUHH, TAU, MINHO) offer well-structured graduate programmes (specifically PhD). They provide courses that can be taken by any DC if they are deemed valuable by the DC and their local Supervisor and are approved by the SB. As a result, any DC will be able to attend post-graduate courses offered by any Network partner. The Network's academic partners, in particular, provide a diverse range of courses in the network's research areas. More particularly, the academic partners of the Network can offer a variety of courses, as follows:

Institution	Academic Programs
UKFLR	Shared facilities with ALU, EQUIP program, classes in molecular medicine, bioethics, biostatistics, materials and microsystems, molecular oncology, and animal welfare.
TCD	Advanced Medical Imaging, Active Implanted Devices and Systems, Medical Device Design Fundamentals, Statistical Signal Processing, Digital Wireless Communications, Next Generation Networks, Internet of Things, Self Organising Technological Networks, Cyber-physical Systems and Control, Optimisation and Control.
UESSEX	Professional practice and research methodology, Network Security, Mobile Communications, AI, Neural Networks and Deep Learning, Programming Embedded Systems, Digital Communications.
ALU	Management, leadership and mental health, orientation and career planning, research practise and teaching qualification, classes in microsystems and information technology, health and biomedical sciences, embedded systems engineering and solar energy engineering.
TAU	Professional practice and research methodology, Commercialization of biomedical devices, standardisation, Classes on Biomedical Informatics and data science, and AI, Classes on micro and nanotechnology, tissue engineering, biomedical engineering, Computational neuroscience, Cellular electrophysiology, Bioelectronics.
SETU	Molecular science and communications, bio-informatics, propagation modelling for intra-body and inter-body communications, AI-aided cell-rovers and cell-painting, AI-aided alternative treatments for palliative care and mental health, Use of Transcriptional Regulatory Network to initiate gene-regulatory treatments.
TUHH	Multifunctional Materials, Materials Physics, Biomaterials, TUHH Graduate and Research Academies (communication & presentation, science writing, career planning).
TUIL	TUIL graduate centre offering a broad soft skill education; Lectures on Biomedical Engineering including signal and image processing, modelling and simulation, inverse problem solutions; International Summer Schools in Biomedical Engineering organised by TUIL (bi-annually).
MINHO	Two main research Centres: The Centre of Physics (functional and smart materials and surfaces for advanced applications) and The Institute for Polymers and Composites (experience in the production of flexible polymeric parts by additive manufacturing).

Courses will be offered in different ways, to increase DC exposure and learning experience, as follows: i. Receptive Classes: The main theoretical aspects will be taught using this method. From the first day, the classes will have a well-defined syllabus, goals, and learning outcomes that will be available to DCs. Before classes, professors will distribute all required materials to DCs (e.g., lecture notes, slides, readings, basic and advanced bibliography) via the project's secure Intranet. This teaching may also make use of "Switched Classroom" instantiations. ii. Interactive Classes: will be based on DCs actively participating in classes and will encourage constructive discussions among DCs as well as between DCs and instructors. The classes will include problem-solving by DCs. DCs will be assigned problems and must think them through beforehand. DCs will also be encouraged to work in groups to solve problems and present their solutions during class. This type of class will help DCs improve their problem-solving, non-traditional solutions, problem analysis,

and teamwork skills. iii. Independent Learning Classes: The classes will also include some sessions devoted to analysis discussion circles, in which DCs will be given scientific materials (slides, published papers, etc.) and asked to study and analyse them before presenting the material in "discussion circles" during class. The DCs are expected to highlight the benefits and drawbacks of the studied material, with a focus on clarifying any doubts that arise from the self-learning process. All classes and courses offered will be of high quality and well-known throughout Europe. DCs will have the option of physically attending courses or participating in video conferences via the project's Intranet. The Supervisory Board will determine the required courses for each DC based on their background and Personal Career Development Plan (PCDP) in collaboration with the local Supervisor(s).

3. Undergraduate/Graduate Project Supervision: All DCs will be encouraged to supervise undergraduate and/or graduate/Master theses or internships (for industrial partners) that are related to their research topics. During their appointment term, each DC will be encouraged to supervise at least one student.

4. Extracurricular Activities: Extracurricular training activities are planned within DC training plans to broaden DCs' technical skills and supplement them with the necessary social and professional abilities. Complementary training activities will include scientific paper writing and presentation skills, which are required for successful research. The Network has a list of courses that will be available to DCs during their appointment period. The length of the courses varies depending on the topic (from 8 to 20 hours). DCs will attend the course in person, but they may use video conferencing on occasion if necessary. Every beneficiary partner will offer either 1-2 courses. All complementary training courses will be closely monitored and approved by the SB. The comprehensive list of planned complementary training courses, the lead institution(s), and their tentative schedule is:

	Title	Lead	M	H	Description
C1	Writing Scientific Papers and Presentation Skills	UESSEX	M8	10h	Organising thoughts and ideas, differences between publications, peer-review, databases, presentation skills, vocal impact and body language.
C2	Commercialization of Biomedical Devices and Software	TAU	M10	10h	Overview of industry requirements, regulations, health technology products, product life cycle, product testing, safety verification, usability standards, ethical issues, clinical evaluation.
C3	Preparing for Pre- and Clinical Trials	UKLFR	M12	10h	Overview of bioelectronic pre-clinical tests in vivo, statistical parameters, open science requirements, phases of clinical trials, statistical bases.
C4	Ethics in a Practical Way	UKLFR	M24	20h	Neuroethical thoughts, group role play, ethical situation simulation, practical training in ethics.
C5	Exploring Novel Tools in Grant Writing Applications	UKLFR	M32	8h	Analysis of grant proposals, team engineering of proposals, peer-assessment, targeted application improvement.
C6	Team Building Experience	ALU	M40	20h	Outdoor challenge, lessons on team building, self-motivation, trust, resilience, leadership skills.
C7	Research Integrity and Impact in an Open Scholarship Era	TCD	M14	8h	Copyright, data protection, data management, scholarly communication, research evaluation, research impact.
C8	Planning and Managing your Research and your Career	TCD	M26	15h	Research process management, time and resource planning, communication skills, career action plan, team impact factors.
C9	Getting Value out of a PhD Beyond Writing Papers	SETU	M34	16h	Proposal writing, securing industrial grants, building start-ups.
C10	Working with Others: Team-Work and Inclusion	SETU	M22	8h	Career management, DEI actions, networking, reputation and esteem.
C11	Hygiene and Safety in Laboratory Activities	MINHO	M10	8h	Laboratory safety, risk assessment, handling of chemical and biological agents, safety procedures, field-work hazards.
C12	Monitoring and Management of Large Projects	MINHO	M18	8h	Project execution, monitoring, control, completion, group discussions, project management techniques, real-world project application.
C13	Research Data Management	TUIL	M11	8h	Data collection, organization, ethics, storage, security, preservation, open data, data management plans.
C14	Biocompatibility and Biosafety of Implants	TUHH	M20	8h	Implant biocompatibility, host responses, testing methods, regulatory standards for medical implants.

5. Scientific Seminars: The BRAINET Network will organise a series of technical seminars, all closely related to the WPs, to deepen and broaden the DCs' technological knowledge on cutting-edge research topics. The seminars will provide all DCs with a comprehensive and holistic state-of-the-art review of relevant BRAINET topics, as well as opportunities for discussion with highly qualified experts. When selecting seminar speakers, the network will consider gender balance. The length of the seminars will vary depending on

the topic (from 10 to 15 hours). A list of the planned seminars at the lead institution/s, and their tentative schedule is detailed below:

	Title	Lead	M	Description
S1	Principles with Neural Interfaces Using Wireless Signals	UESSEX	M7	From brain implants to interconnected neural networks, exploring wireless signals in neuroscience.
S2	Bioelectronics Concepts for Neuronal Interfaces	TAU	M9	Principles of bioelectric signal sources, neuronal stimulation, and computational models of neuronal systems.
S3	The Foreign Body Response from Brain Implants	UKLFR	M11	Insights into the body's immune reactions to traumatic brain injuries and neural implants.
S4	Information Processing for Cyber-physical Systems	SETU	M21	Self-developing cyber-physical systems in health care, including networked infrastructures and interacting robots.
S5	Complex Systems for Intra-Body Networks	TCD	M15	Future networks as ultra-large-scale, dynamic, complex systems with autonomous network frameworks.
S6	Alternative Computational Methods for Future Networks	TCD	M17	Use of Artificial Intelligence in the telecommunication industry for future network analysis and design.
S7	Introduction to CMOS Circuit Layout and Its Impact	ALU	M19	Difference between circuit design and layout, layouting concepts, and separation of digital and analog circuits.
S8	Multidisciplinary Optimization Methods	SETU	M21	Various optimization concepts including numerical, statistical, learning-based, and evolutionary optimization.
S9	Nanoarchitected Thin Films: applications to materials surface functionalization	MINHO	M27	Preparation and characterization of nanoarchitected thin films for tailored material surfaces responses.
S10	Temporal Interference Stimulation	TUIL	M13	Basics of electric brain stimulation, including invasive and non-invasive approaches, and stimulation technologies.
S11	Multifunctional Materials	TUHH	M33	Fabrication, performance, and bio-compatibility of multifunctional materials for energy storage and conversion.

Additional seminars and invited talks: The above list of seminars is not intended to exclude further events, for example at conferences organised by partners or otherwise. In addition to public seminars, worldwide experts in topics related to the BRAINET Network will be invited to give talks to DCs, also inviting all associated partners to host seminars either online or in-person, with preference for online events. We will work towards a goal of bimonthly seminars.

6. Summer/winter schools: Incorporating summer/winter schools into the BRAINET project will not only facilitate focused learning and skills development but also accelerate collaboration, networking, and continuous engagement with the project's evolving aspects, thereby enriching the overall research and development process. Here is our list of planned summer/winter schools:

	Title	Lead	M	Dur	Description
SS1	Electrophysiologic Recordings from Living Sources for Engineers	UKLFR + ALU + TUHH	M20	3 days	Comprehensive understanding of neural electrophysiology, practical applications in biomedical engineering, ethical considerations.
SS2	Introduction to Wireless Intrabody Communications	TCD + SETU + UESSEX + ALU	M13	3 days	Fundamentals of intrabody communication, signal propagation, communication protocols, applications in medical devices and neural interfaces.
SS3	Electric Interference Stimulation	TAU + TUIL	M33	3 days	Technical skills in biomedical engineering, neuroscience methods, biomedical signal processing, design of HW and SW for brain stimulation.
WS	Additive Manufacturing: From Lab to Factory Floor	MINHO	M24	3 days	Methods and technologies in additive manufacturing, high-speed processes, improved materials, optimization software, experimental component.

7. Integration within Organized Research Teams: Every Network partner is a well-established organisation with well-structured teams and groups. The recruited DCs will be encouraged to participate fully in all team activities and organisational tasks within their hosting organisations in order to gain experience in organisational structure, operation, management, and so on. All partners hold regular internal meetings (at least once a month) to discuss accomplishments as well as current issues of various groups (e.g., teaching, theses, funded projects, proposals, technology transfers in the case of academic partners or funded projects, proposals, commercial area, implementation team, public relations in the case of industrial partners). Each DC, whether at their hosting or secondment institute, will be encouraged (and in some cases, required) to attend those meetings in order to increase their awareness and knowledge of the structure and operating environment of various organisations. We will also host a team building event, near completion of the recruitment process, so that supervisors and DCs meet in-person as a team for accelerating capacities of the network towards developing a community.

8. Participating and contributing to funded research projects: Each DC will have numerous opportunities to follow and participate in funded research projects, which will serve as an additional source of training. Currently, all Network partners have funded projects (internal, national, etc.). Following various funded projects and interacting with researchers involved in the funded projects will benefit the DCs. DCs will be kept up to date on project planning, management, and execution. They will also be permitted to attend meetings of ongoing European projects, which will provide them with valuable practical experience. To ensure coherence with the training programme and interest in the DCs' career, the SB must approve which projects each DC participates in. Currently, the Network partners are involved in several research projects (see section 5), related to BRAINET.

9. Attending internationally organised conferences and workshops: As the main component of DCs' complementary training, all DCs are required to achieve a minimum number of publications, and will thus attend and participate in international gatherings of the world's top experts in their fields. Aside from the benefits of disseminating their findings, DCs will have the opportunity to meet other researchers working in the same field, be exposed to new ideas addressing the same or similar problems, and discuss their findings with other researchers. DCs are also encouraged to publish in relevant scientific magazines and journals with a high impact factor. At the start of the project, a list of targeted conferences, workshops, journals, and magazines will be prepared. The SB must approve the list.

10. Organization of Workshops and Dissemination Days: A workshop, meant as a showcase event, within an IEEE conference and a final conference will be organised by the Network throughout the lifetime of the project. During those events, DCs are expected to present their progress and major accomplishments. All events will provide excellent opportunities for DCs to meet, learn about, and discuss the research and progress of other users. The Network also intends to hold three Transectorial Dissemination Days (TDDs) dedicated to the presentation of DCs' results of interest to industry and with close ties to the IEEE UK-Ireland branch (Prof Marchetti and Dr Dey are active members). The goal of these TDDs is for all DCs to gain experience in presenting their work to industrial stakeholders. All events will be open, inviting young researchers, local industry, and experts from Network companies and standardization groups to provide feedback. This input will be vital for advancing DCs' research, with DCs encouraged to implement the suggestions.

11. Periodical plenary meetings (2 per year): DCs will be invited to Network Plenary Meetings and encouraged to participate in WP discussions. DCs will also be exposed to general discussions about the project's overall progress, as well as specific discussions about the individual WPs in which they are involved. The plenary meetings serve as a forum for presenting comprehensive results from DCs in various WPs, exchanging ideas, and comparing results and methods. Special emphasis will be placed on the fruitful discussion of the results, which will be open to all Network members in order to provide a more multidisciplinary environment and feedback. The DCs will gain experience and learn more about how a project is run, managed, and organised, as well as how decisions are made within collaborative projects, through the Plenary meetings. WP leaders should request that DCs involved in their respective WPs present their completed results.

12. Secondments (training activities) Within their PCDP, DCs will spend a training period of 6 months in the premises of another partner in the Network. The DCs are also expected to pay short visits to other partners if the SB sees that as necessary. The institution, where the DC spends their secondment, has been chosen based on the research topics of the DC. The goals of the planned secondments (visits) are i) strengthen the collaboration among partners of the Network. ii) foster knowledge exchange, iii) acquire the required knowledge based on the needs of research of the DC, and iv) expose the DC to different research environments (from academia/research centre to industry and vice versa). The secondments will be precisely planned, and the schedule and details of the secondment of each DC will be included in their PCDP. Industrial partners will host DCs and enable their own DCs to undertake secondments with other partners, with plans approved by the SB. All partners actively contribute to DC training, with the SB playing a vital role in its success throughout the project. A tentative schedule is shown in Section 3.1 within each individual project.

1.4 Quality of the supervision

The network consortium has been meticulously crafted with the overarching objective of delivering training of the highest quality, encompassing theoretical research, exposure to market-driven research, prototype experimentation, and the imparting of high-value transferable skills. The consortium consists of 9 academic and industrial partners, all esteemed institutions committed to providing expert guidance to Doctoral Candidates (DCs) in their respective research domains.

1.4.1 Qualifications and supervision experience of supervisors

Qualifications of Supervisors - All potential supervisors within the consortium have been selected based on their exceptional qualifications, expertise, and standing in their respective fields. The supervisors are highly qualified professionals, many holding the academic rank of professors, ensuring a strong foundation

of knowledge and experience. **Experience in Supervision** - The network partners, both in academia and industry, bring extensive experience in supervising research and training early researchers, including the supervision of Master's and Ph.D. theses. This wealth of experience ensures that DCs receive mentorship from seasoned professionals capable of guiding them through the complexities of theoretical research and market-driven projects. **International Recognition and Roles in Professional Organizations** - Supervisors, backers, and researchers associated with the project have achieved international recognition for their contributions to research and have served in prominent roles within prestigious organizations such as IEEE (Editor in Chief, editors, Technical Program Committee members, etc.), EAMBES (European Alliance for Medical and Biological Engineering & Science), IFMBE (International Federation for Medical and Biological Engineering), and the FENS (Federation of European Neuroscience Organizations). **Publication Record** - Each supervisor boasts a respectable list of publications, showcasing their scholarly contributions. This includes patents, journal articles, books, and conference presentations, as evidenced on their individual ORCID profiles. **Professional Development and Training** - In adherence to the European Charter for Researchers, all potential supervisors have agreed to serve as mentors to DCs. To enhance their capabilities, supervisors undergo annual refresher courses and receive specialized training designed specifically for supervision. The coordinating institution offers its supervisor training and professional development courses, fostering an exchange of best practices and ensuring a consistently high standard of mentorship across all consortium institutions. **Capacity Building and Mentor Pairing** - The program is committed to increasing institutional capacity by strategically pairing less experienced advisors with more seasoned mentors, providing a balanced and effective approach to mentorship. **Time Allocation and Monitoring** - Each school within the consortium ensures that supervisors have a defined amount of time dedicated to their mentees, following their individual policies.

1.4.2 Quality of the joint supervision arrangements

The hybrid supervising expertise encompasses wireless and microelectronic experts, computer scientists, material scientists, and neuro-engineers, providing DCs with a rich and diverse supervisory environment. **Supervision** - DCs benefit from the combined perspectives of both a local supervisor at the hosting organization and a co-supervisor from another organization, ensuring intersectoral and international supervision. The inclusion of experts from different sectors and organizations broadens the scope of DCs' training, exposing them to diverse perspectives and methodologies. **Supervision Time Allocation** - At least fifteen percent of each supervisor's time is dedicated to monitoring the work of the DCs, reflecting a commitment to effective guidance and mentorship. **Systemic Approach to Supervision** - The Steering Board (SB), ensures that DCs across the Consortium perform at an acceptable level, leveraging the collective expertise and experience of the SB members. **Personalized Career Development Plans (PCDP)** - Each DC, in collaboration with their supervisors and the training coordinator, develops a Personalized Career Development Plan (PCDP) tailored to their individual interests and goals. The PCDP follows the respective graduate school requirements and serves as a roadmap for the DC's research and training. **Supervisory Committee (SC)** - Each DC is supported by a three-person supervisory committee (SC), consisting of a supervisor, a co-supervisor, and a "6-month rotating" member from other APs. The SC ensures that the PCDP is established and regularly reviewed, typically on an annual basis, providing a structured mechanism for ongoing assessment. **Regular SC Meetings** - SCs report the DC's activities to the relevant authority, fostering transparency and accountability. Monthly meetings between DCs and their SCs facilitate continuous supervision of research and training, allowing for timely adjustments and support. **Plenary Sessions and Video Conferencing** - Plenary sessions provide dedicated time for SC meetings, ensuring a collective and collaborative approach to supervision.

#§REL-EVA-RE§#

2 Impact#@IMP-ACT-IA@#

2.1 Contribution to European doctoral training and innovation capacity

Brain diseases impose significant socio-economic burdens, and Deep Brain Stimulation (DBS) offers a valuable treatment option for several. BRAINET addresses this by training a new generation of engineers, focusing on minimal invasive wireless neuronal interfaces for semantical and interference brain stimulation. The collaborative project draws expertise from multiple European organizations, aligning with European initiatives³⁵. European collaboration in BRAINET leverages integrated capabilities and resource sharing for impactful projects. Involving industrial partners, R&D facilities, and associated partners promotes consistency in European research, showcasing achievements. Training researchers in minimal-invasive stIS contributes to advancing wireless brain-computer interfaces, ensuring Europe's leadership in science and medical technology globally.

³⁵<https://spanish-presidency.consilium.europa.eu/en/news/leon-declaracion-european-neurotechnology-human-rights/>

2.1.1 Meaningful contribution of the non-academic sector to the doctoral training

Facilitation of Standardisation Activities : Associated Partners (APs) play a pivotal role in enhancing various standardisation activities within the project. Their involvement can extend to providing memberships or covering expenses for academic partners, enabling them to actively participate in standardisation events and contribute valuable insights.

Support for Patent Submissions : APs can actively contribute to the patent submission process by providing financial backing for filing patents. Additionally, their expertise, particularly from financial and legal departments, can prove invaluable in navigating the intricacies of patent-related matters.

Indirect Contributions through Collaborative Training : Industrial partners contribute directly and indirectly by closely collaborating with academic institutes in the training program. Planned secondments involve DCs being sent to industry and other sectors, facilitating the transfer of cutting-edge knowledge and steering academic research towards commercial orientation.

Impact on Organizations and Market Opportunities : The program can significantly impact on participating organizations, fostering richer contributions and opening industrial contacts. It also provides insights into the medtech landscape, patient perspectives, and identifies new project opportunities, fostering a comprehensive understanding of market needs.

Strategic DC Utilization for Future Hiring : Participating organizations can benefit from the DCs for future hiring purposes. Having explicit knowledge of this Doctoral Network BRAINET allows partners to identify candidates who align best with their organizations, minimizing hiring risks.

Exposure to Regulatory Realities : With recent EU Medical Device Regulations, academic institutions may lack exposure to regulatory issues. APs bridge this gap, exposing DCs to real-world valorization and certification processes, enhancing their employability and understanding of regulatory realities for potential employers.

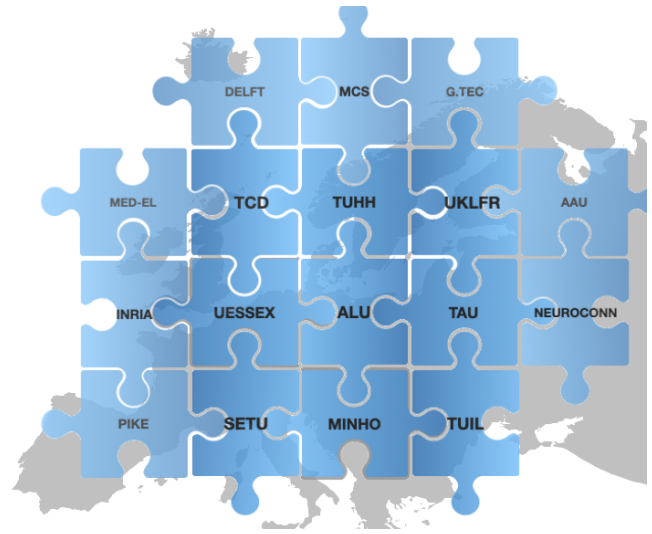


Figure 4: Tapestry of the BRAINET Consortium on Europe

2.1.2 Developing sustainable elements of doctoral programmes after the end of the funding

☛ **Sustainability of Training Programs and Transferable Skills Training :**

- a) Open Access* - BRAINET will ensure that the training programs and transferable skills training materials are made open-access or available for a fee to external doctoral students and institutions. This can be done through online platforms, webinars, or open educational resources.
- b) Establish an Endowment* - BRAINET will develop graduate and doctoral programs through funding schemes like MSCA DNs, collaborating with the business community, including SMEs. Co-developed programs better suit employers' needs, and DCs can later participate in future networks as co-supervisors. Industry involvement ensures academic programs adapt to evolving market needs.
- c) Developing SME-academic networks* - BRAINET utilizes national support, e.g. Enterprise Ireland funding, to foster informal networks and activities between SMEs (APs) and academia (Beneficiaries). These networks address local needs, enhancing inter-sectorial mobility through meetings, events, conferences, job fairs, and more. Involving SMEs is facilitated by national industry federations, representative organizations, and SME clusters or
- d) Training supervision* - BRAINET offers DCs the chance to work with dual steerings, one from academia and another from industry, fostering collaboration. It also prepares industry-based Experienced Researchers for supervision roles through ad-hoc training and accreditation by academic professionals.

☛ **Sustainable Cooperation and Collaboration :**

- a) Collaboration and Research Network* - In line with the objectives of energy-efficient Europe-flagship (www.eea.europa.eu/policy-documents/a-resource-efficient-europe-flagship), BRAINET leverages beneficiaries' work to create a vast research network, fostering stronger industry-research links and providing skills to partners. It supports European Research Area expansion, encourages global talent attraction, aids sustainable development in developing countries, and promotes a culture of innovation. The collaboration addresses key issues in communication systems, health science, neurotechnology, deep interference simulation, and data
- b) Alumni Networks* - BRAINET will establish a federated alumni network for graduated doctoral students, facilitating ongoing collaboration and information sharing. This platform aims to explore funding opportunities for joint research projects, considering grants from government agencies, private foundations, or industry to support continued collaboration.

☛ **Sustainability of Researcher Recruitment :**

- a) Adherence to Code of Conduct* - BRAINET

commits to adhering to the code of conduct for the recruitment of researchers even after the project ends. The PIs from the beneficiaries will ensure that the principles of fairness, transparency, and equality in recruitment are integrated into the partner institutions' policies and practices. **b) Review and Adapt Policies** - BRAINET commits to periodic reviews and updates of DC recruitment policies to align with the code of conduct, adapting to evolving legal, societal, and educational landscapes. This ensures ongoing effectiveness and provides opportunities to identify and address weaknesses, fostering continuous improvement in recruitment practices. **c) Diversity and Inclusion** - BRAINET prioritizes diversity and inclusion in recruitment practices, aligning with the code of conduct. This involves actively seeking candidates from diverse backgrounds, avoiding unintentional barriers, providing cultural competency training for staff, and offering mentorship to promote success for students from underrepresented groups. **d) Data and Reporting** - BRAINET consortium will establish mechanisms for collecting and reporting data on recruitment practices to ensure compliance with the code of conduct and to track progress over time. The beneficiaries will be using the data to set goals and track progress over time, which ensures that the beneficiaries are making meaningful strides toward inclusivity and fairness for further recruitment of DCs.

2.2 Measures to enhance the career, employability and skills development of researchers

The BRAINET project and its comprehensive training program have a profound and multi-faceted impact on the career prospects of the Doctoral Candidates (DCs), preparing them for successful careers in both academia and the industry. Here's how this initiative equips the DCs with technical and transferable skills that significantly enhance their employability: **Technical Excellence** : BRAINET aims to enhance DCs expertise in communications, nanomaterials, and neural interfaces, making them highly attractive to academic institutions and private sector companies seeking professionals with cutting-edge technical skills. **Multidisciplinary Exposure** : Through a multi-national, multi-disciplinary research program, DCs gain a holistic perspective bridging traditional academic boundaries, broadening their knowledge and skills. This adaptability is highly sought-after in academia and industry. **Publication Skills** : The training program incorporates courses and seminars to enhance DCs technical writing skills, providing real-world experience through submissions to reputable conferences and journals. This sharpens communication abilities and enhances credibility in their respective fields. **Participation in High-Impact Conferences and Training Courses**: DCs receive network-funded opportunities to attend prestigious conferences and training courses, staying abreast of the latest research and industry developments. This positions them as valuable assets in academia and industry. **Personal and Presentation Skills** : DCs develop critical personal skills, including presentation skills, through offered seminars and courses. This enhances their employability and ability to communicate research effectively in job interviews, academic presentations, industry pitches, and public speaking. **Multi-National and Multi-Disciplinary Collaboration** : DCs will gain experience integrating within diverse teams, fostering efficiency and adaptability in group settings. They will be encouraged to work both independently and take initiative, essential qualities for a successful research career in academia and industry. **Research Management Exposure** : Involvement in management tasks, meetings, and administrative issues at partner institutions will ensure DCs are well-versed in research project oversight and organisational aspects. This experience enhances their ability to take on leadership roles in academia or industry. **Networking with High-Level Experts** : DCs will interact with leading experts in their field through the consortium's network and top conferences, providing opportunities for collaboration, mentorship, and potential employment. **Research Funding and Entrepreneurship Skills** : DCs gain hands-on insights into research funding applications, proposal writing, and the practicalities of operating new businesses and creating start-ups. This broad skill set widens their opportunities to apply their expertise by choosing the entrepreneurial path or collaborate with industry experts.

BRAINET offers DCs a unique platform for a skill set that blends technical excellence with transferable skills. This combination is attractive to employers seeking well-rounded professionals capable of groundbreaking research, effective communication, management, and collaboration. DCs are equipped to shape the future of bioelectronic medicine and biomedical engineering, positioning them as motivated and successful professionals ready for impactful careers in various settings. Particularly, BRAINET actively supports DCs' exploration into the opportunities of entrepreneurial spin-outs, given that several PIs are serial founder and mentors to start-ups. Based on the high interdisciplinarity and reach-out goals of BRAINET, DCs might even consider a career in science communication and consultancy.

2.3 Suitability and quality of outcomes and impacts, dissemination and exploitation plan#@COM-DIS-VIS-CDV@#

2.3.1 Plan for the dissemination and exploitation activities, including communication

🔗 **Dissemination Strategy** - *Website Development and Maintenance (M1 and ongoing)* - We

will establish a secure internal server for document sharing; maintain an up-to-date public-facing website for external dissemination; regularly update with public information, including announcements, workshops, tutorials; use the website to share accepted public deliverables. **Conference and Journal Publications** - The consortium identifies top scientific journals, magazines, and conferences; actively disseminates results through selected publications (e.g., IEEE/ACM Trans. Networking, Nature) with a minimum of two submitted full journal papers submitted per DC and two conference papers (e.g., Infocom, Globecom, IFMBE MBEC, IEEE NES, MEDICA); showcases test-beds at dedicated events; ensures joint publications among partners; updates the list based on SB feedback if needed. **Open Access Policy Compliance** - We will adhere to Horizon Europe's open access policy; use an open access repository for publications and bibliographic metadata; include required information from the European Commission; provide a unique repository number in project reports. **Contributions to Standardization** - We will actively contribute to standardization in relevant fields; engage with standardization organizations to incorporate project findings; disseminate information about contributions through the project website and relevant publications. **Workshops and Dissemination Days** - Organize workshops and dissemination days to share achievements with a broader audience; invite key stakeholders, researchers, industry experts, and patient's societies; use these events for in-depth discussions and networking opportunities. **Interaction with Scientific Communities and Patient Initiatives** - Foster participation with scientific communities (e.g. IFMBE, EAMBES, FENS, VDE) and improving patient engagement (e.g. EFNA, European Patients Forum) related to project objectives; participate in relevant forums for insights and feedback; establish collaborations & partnerships with other research projects to enhance collaboration & knowledge exchange. **Feedback Mechanism and Plan Updating** - Establish a feedback mechanism involving the Steering Board and stakeholders; regularly assess and update the dissemination plan for improved effectiveness.

☞ **Plan for exploitation** - BRAINET aims to significantly influence wireless neural interfaces and bio-electronic medicine, with collaborative efforts from universities, research institutes, and industrial entities maximizing project outcomes aligned with their objectives. **The universities within the consortium are strategically positioned to leverage the project results to enhance their research capabilities, training programs, and overall knowledge base towards new opportunities in commercialization.** Through each university's transfer offices and innovation HUBs, BRAINET plans exploitation through generation of new IPR, as well as the eliciting of new start up activities. Through our training program DCs will have the opportunity to work with PIs and companies to drive wireless brain interfaces in multiple but yet more specific means of innovation. In order to early expose the DCs to the mindset needed to spin-off a company from the lab bench, an inspiring simulation (event 37) will be offered, involving experienced entrepreneurs and associated partners. DCs will be encouraged to develop startup ideas related to their work and participate in according competitions like SPARK in Finland (sparkfinland.fi/) or EXIST in Germany (www.exist.de/).

BRAINET aims to extend the project's impact beyond academia, catalyzing advancements resonating with industrial, clinical and patient interest group's imperatives.

☞ **Communications** - Together with the Public Relation Offices in our organisations BRAINET outcomes will provide a legacy in wireless brain interfaces that need to be communicated to all mobile ecosystem stakeholders, including the general public, through effective public engagement and effective communication. To accomplish this goal, we have devised the following agenda: **Social Media Presence** - BRAINET acknowledges social media's role in promoting scientific endeavors; we assign DCs to maintain updated profiles on platforms like LinkedIn or federated microblogs; establish social media accounts alongside the project website in Month 2 for a comprehensive online presence. **Project Leaflets** - BRAINET annually creates public-friendly advertising leaflets highlighting program achievements in biomedical engineering, updating content regularly to reflect significant milestones and progress. **Multimedia Material** - We produce and distribute multimedia material, featuring videos of practical experiments and real-time simulations, disseminated through the project website and relevant platforms to enhance accessibility; initiate material production in WP1, releasing content after each milestone. **Publicity Materials** - Develop high-quality publicity materials (brochures, infographics, press releases); distribute to research communities, special interest groups, and stakeholders; ensure materials are tailored for academic and non-academic audiences. **Articles in the Press** - Generate news articles about BRAINET's advancements for local newspapers, emphasizing Marie Curie Actions' benefits and positive project impacts; integrate press articles into WP1, aiming for at least one from each partner throughout the project. **Activity Days** - We Organize local activity days, led by academic partners, engaging university students and introducing project research; schedule strategically at M21, M28, M35, M40 to align with milestones. **Public engagement** We encourage DCs and PIs to visit local schools, organize public talks (e.g. Cafe Scientifique, Pint of Science, Freiburg), science nights (e.g. Science Slams), science week (e.g. Brain Awareness Week) to foster interest and public engagement. Thus, we support all

partners not only to present research results, fostering interaction with the academic community, but leave the academic ivory tower and engage with the public:

Activity & Details	Partner	Target Group	Audience Size	Month	WP
Peer-Reviewed Articles	All	Research Community	30 conf, 15 journals, 20 pre-prints	Start in M12	WP2-5
Patents and IPR	All	Industry	6 Pat.Pend.	Start in M12	WP2-5
Social Media Presence	UESSEX	Research Community, Industry, General Public	20k views, 1500 flws, 200 bi-wkly toots	Start in M2	WP1
Project Leaflets	SETU/TCD	General Public, Research Community	2k prints, 5k dwnlds	Annually	WP1
Multimedia Material	ALU/MINHO	General Public, Students, Research Community	10k views	After Each Milestone	WP1
Articles in the Press	TUIL/TAU	General Public, Local Communities	25, 60 press releases	Throughout Project	WP1
Activity Days	ALU	University Students, Academic Community	500 students, 50 academics	M21, M28, M35, M40	WP2
General Communication and Public Engagement	All Partners	General Public, Research Community	1000 pub goers, 5000 patients, 5 national patient societies	Ongoing	All WPs

2.3.2 Strategy for the management of intellectual property, foreseen protection measures

BRAINET's Intellectual Property Rights (IPR) regulation is detailed in the Consortium Agreement (CA), aligning with EC and national regulations. **IPR management strategy is set by the project internal management board, which also will resolve conflicts of interest, also highlighted in Task 1.4.** This mandatory agreement is formally accepted and signed by all partners before project commencement, serving as a legally binding framework for BRAINET's IPR matters, with key aspects articulated as follows: **a) Background Knowledge Ownership** : Any background knowledge brought into the BRAINET project by individual partners will be owned by the respective partner introducing it. The CA will explicitly specify the parameters under which other partners can utilize this background knowledge. **b) Foreground Knowledge Ownership and Access** : Knowledge generated during the project (foreground knowledge) will be jointly owned by the partners responsible for its creation, as per CA rules. All partners will have access to this knowledge during the project duration. However, post-project, the CA will define the rights of each partner to utilize this knowledge. **c) IP Protection and Exploitation** : Prior to the publication and dissemination of research results, partners developing new knowledge will be granted a reasonable timeframe to protect their Intellectual Property. Subsequently, after due consideration of IPR issues, the new scientific knowledge will be made public, contributing to the benefit of Europe and its citizens. **d) Favourable licensing conditions for spin-outs** : Partners and particularly DCs seeking to use BRAINET's IPR for spinning out a start-up will receive favourable licencing agreements as set out in the CA. **e) Promotion of IPR Helpdesk** : Partners and individual participants seeking guidance on IPR protection will be encouraged to utilize the IPR Helpdesk (www.ipr-helpdesk.org). This resource aims to provide assistance and clarification on questions related to Intellectual Property Rights, ensuring a comprehensive understanding among project participants.

#§COM-DIS-VIS-CDV§#

2.4 The magnitude and importance of expected scientific, societal and economic impacts

The proposed measures aim to boost European researchers' capabilities, fostering global R&I in bioelectronic medicine, wireless neural interfaces, semantic interference stimulation, neural implant microelectronics, intra-body communications, and smart implantable electrodes. Anticipated breakthroughs in Deep Brain Stimulation (DBS) research are projected to drive a significant global market expansion from \$3.5 billion in 2023 to \$9.2 billion in 2033³⁶. The evolution of neural engineering, facilitated by wireless technology, introduces dynamic tools for brain activity stimulation and recording. Addressing clinical challenges involves creating in-silico models, expertise from diverse engineering fields, real-time data access, and navigating regulatory compliance. Successfully managing these challenges is crucial for deploying advanced technologies effectively in clinical settings, ensuring efficacy and regulatory adherence.

2.4.1 Expected scientific impact(s)

Understanding and Treating Brain Disorders: BRAINET aims to pioneer novel brain disorder treatments through integrating interference stimulation in DBS. This holistic approach promises a deeper understanding and effective management of neurological disorders, potentially advancing neurotherapeutics significantly. Interference stimulation adds focal points, steerability, and modulation advantages to conventional DBS and while integrated in a wireless system, it benefits from the minimization of patient hurdle with implants. **Innovations in Nanostructured Materials**: The project aims to revolutionize neurology and

³⁶TrendLens Insights, "Global Bioelectronic Medicine Market Opportunities and Trends". Report ID : 1297431 | November 2024
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brain-machine interface fields by creating new nanostructured materials for implantable devices traversing the skull. Using polymer and graphene nanotechnology to make neural devices last longer and work better with the body could lead to more effective and long-lasting devices, which will push the limits of neuroprosthetics and neural augmentation. **Targeted Deep Brain Stimulation:** Spatio-temporal Interference Stimulation allows to adjust the targeted brain region without being fixed to a pre-defined implant spot. This clearly pushes the limits of neuroprosthetics and neural augmentation. **Revolutionizing Wireless Implantable Devices:** BRAINET's creation of a semantic multiphysics wireless platform changes the way implantable devices work by making them more reliable, flexible, and able to block interference, which leads to better patient outcomes. Pure magnetic power-transfer already started to be a gamechanger in the world of body implants. **Advances in Body-Coupled Networking :** BRAINET advances body-coupled networking with ad-hoc protocols for WTBS. This significant stride ensures effective communication among devices, ushering in advanced neural networks for complex neurological challenges. *In essence, BRAINET's multi-faceted approach holds the promise of groundbreaking advancements across various fronts, positioning it as a pioneer in the quest to address and transform the landscape of neurological research and treatment.*

2.4.2 Expected economic/technological impact(s)

Parkinson's Disease (PD) treatment as a prominent example for other neurological diseases costs the European healthcare system over 200 billion Euros annually. Per patient, costs associated with the DBS itself run typically around 170k euros per patient, not adding their transportation to clinics, adding several 1k euros per year. With an aging population, the urgent need for safer and more effective treatments of neurological disorders aligns with BRAINET's overarching goal to address these challenges and improve long-term disease management. *We predict that BRAINET vision can decrease the costs of DBS between 30%-50%.*

BRAINET aims to revolutionize the treatment of neurological disorders, addressing significant healthcare issues, and acknowledging economic impact. The project foresees a transformative boost in the electroceutical market, identified by McKinsey as a key sector in the innovation of medical treatment.

The BRAINET technology holds the promise of becoming a major player in these markets, and special attention will be dedicated to identifying innovations and protecting Intellectual Property Rights (IPR). Multiple startups will be encouraged, DCs motivated to consider spin-outs via incubator-type entities and preferred technology licensing. This approach fosters innovation, job creation, and economic growth, expanding neurological healthcare solutions across diverse industries in the EU.

2.4.3 Expected societal impact(s)

DBS is explored for various neurological conditions, but its use is limited due to risks. BRAINET's networked, wireless, transcalvarial, non-penetrating brain stimulation (WTBS) is a revolutionary technology poised to broaden application, potentially reducing disability rates. Unlike DBS, WTBS' semi-invasive profile spans conditions like Parkinson's and depression. While DBS is reserved for severe cases, WTBS holds promise for widespread impact, improving outcomes across diverse neurological disorders, making a significant societal and economic contribution by enhancing the lives of a broader patient population.

BRAINET positions Europe at the forefront of the next generation of DBS development and clinical markets, especially with the electric field interference introduced by networked WTBS technologies.

Beyond clinical applications, BRAINET envisions influencing public understanding and acceptance, recognizing that dissemination of technical information, philosophical and ethical analyses, and engagement with stakeholders are vital for clinical impact. Ethical considerations, emphasized in WP7, are vital for public acceptance, particularly in brain-related innovations. Public opinion will guide the design of WTBS products post-BRAINET, aligning innovations with societal values. This approach aims to advance medical technology ethically, ensuring socially responsible and widely accepted progress in neurological treatments. #§IMP-ACT-IA§#

3 Quality and Efficiency of the Implementation

#@QUA-LIT-QL@# #@WRK-PLA-WP@# #@CON-SOR-CS@# #@PRJ-MGT-PM@#

3.1 Quality and effectiveness of the work plan, risks and work packages

3.1.1 Work Package (WP) List description #§CON-SOR-CS§# #§PRJ-MGT-PM§#

The network's work plan is divided into four parallel tracks: management (WP1), research (WP2-5), training (WP6), and dissemination (WP7). This section defines the structure of the various WPs, including deliverables and milestones that will be used to monitor and ensure the proper progress of each WP and the project as a whole. WPs have a month-to-month schedule.

Table 3.1 a Work Package (WP) List

WP No.	WP Title	Lead No.	Start M	End M	Type	Lead	Researcher involvement
1	Project Management and Coordination	1	M1	M48	Management	UKLFR	All
2	Wireless Transcalvarial Brain Stimulator Technology	4	M4	M44	Research	ALU	All
3	Semantical and Multiphysical Implant Communication Networking	3	M4	M44	Research	UESSEX	UESSEX, TCD, SETU, UKFLR, ALU, TUHH, MINHO
4	Preparatory Work of Semantics-based Interference Stimulation	7	M4	M44	Research	TUIL	TUIL, MINHO, TUHH, TAU, UKFLR, ALU
5	Integration and Demonstration of Transcalvarial Networked Brain Stimulators	8	M12	M44	Research and Innovation	TAU	TAU, UKFLR, UESSEX, ALU, TUHH, MINHO
6	Networking and Training	2	M1	M48	Training	TCD	All
7	Communication, Dissemination, Exploitation, Regulations and Standards	5	M1	M48	Communication	SETU	All

3.1.2 Description of Work Packages

Table 3.1 b Description of Work Packages

WP Number	1	M1-M48
WP Title	Project Management and coordination	
Objectives: WP1 will run throughout the duration of the project, will set forth all the management and technical coordination activities required for a smooth running of the project. The main goal is the overall management of the work, in order to reach the project objectives within the agreed budget and time scales. Description of Work and Role of Specific Beneficiaries / Associated partners: ➤ Task 1.1 - General project management (Lead: UKLFR, Others: ALL; M1-M48): Coordination of the consortium, providing an accurate control of the project enabling the partners to achieve all objectives of the project on time and budget, and with a high-quality level. Consortium wide meetings will happen at 12, 30 and 48 months, supported by monthly virtual meetings to ensure productivity and continuity. We include meetings and management reporting. ➤ Task 1.2 - Technical Management (Lead: UKLFR, Others: ALL; M1-M48): Technical work groups will be established and managed to coordinate and unite hardware, software, and wetware cross cutting WPs by bi-weekly virtual and if needed short scheduled personal meetings. These will be augmented by “educational council meetings” dealing with training issues and soft skill developments of DCs including secondment positioning. ➤ Task 1.3 - Quality Assurance and Risk Management (Lead: UKLFR, Others: ALL; M1-M48): Risks and opportunities identified in the outset of the project will be monitored and the associated contingency plans will be respectively adopted because risks in this project are very high. ➤ Task 1.4 - Innovation Management (Lead: TCD, Others: ALL; M1-M48): Based on the ground breaking type of work done in this project, valorization of research will be particularly emphasized. Jointly with all WPLs, strategies for managing the IP, technology transfer and exploitation activities related to project results will be devised and undertaken with all relevant steps. ➤ Task 1.5 - Data Management Plan (Lead: UESSEX, Others: ALL; M1-M48): Describes the data collected across all Work Packages (WPs), the data collection methods, actions to secure data, and actions to secure FAIR (Findable, Accessible, Interoperable, Reusable) data.		
WP Number	2	M4-M44
WP Title	Wireless Transcalvarial Brain Stimulator Technology	
Objectives: Based on an initially identified biophysical relevant specifications microelectronic hardware will be developed, which will provide (i) Central command unit, (ii) Communications stack, (iii) Environmental sensing unit, (iv) Physical communication layer, (v) Energy production, (vi) Advanced tissue coupling. Description of Work and Role of Specific Beneficiaries / Associated partners: ➤ Task 2.1 - Requirement Gathering and Design (Lead: UESSEX, Others: ALL; M4-M18): <i>This tasks activates the Technical work groups (Task 1.2) to identify hardware, software, and wetware specifications by proven online and in-person methods. A thorough literature collection is integrated to the specification and design requirements for a WTBS system according to biological (e.g. biocompatibility and trauma-avoidance), electronic (e.g. CMOS process and heat dissipation) and biophysical (e.g. signal spread and inhomogeneity) boundary conditions. Initial overall cross cutting technological design definitions are made here. Methods from interactive and agile design are to be used.</i> ➤ Task 2.2 - Biophysical brain characterization for wireless distributed interfaces (Lead: UESSEX, Others: ALL; M4-M18): We identify the most relevant biophysical parameters (e.g. inhomogeneous mechanical and electrical brain properties) and biological confounds (e.g. maximal permitted energy dissipation and permitted current densities) for the intended application. ➤ Task 2.3 - Power and information transfer technology (Lead: TUHH, Others: SETU, TCD, UESSEX, ALU; M4-M44): We will be dedicated to the fabrication and optimization of materials that convert in a wireless manner magnetic energy outside the tissue (brain) via mechanical deformation energy to electrical energy. This electrical energy can then be used for communication, sensing and stimulation. To achieve wireless coupling of magnetic energy to electrical energy, a PPy-Si material will be combined with a magnetostrictive material, e.g. Metglas. ➤ Task 2.4 - CMOS-integratable technologies for neural interfaces and brain stimulation devices (Lead: ALU, Others: ALL; M4-M44): Development of CMOS-integratable stimulation strategies for local WTBS devices, seeking to optimise and refine stimulation methods. Including CMOS-based communication and synchronisation between multiple WTBS units, ensuring efficient and harmonised operation across different devices. We will design, laying out, and fabricate a bidirectional neural interface using CMOS technology, encompassing all stages from initial design to final production. An architecture development for a neural-data pre-processor, a crucial component for interpreting and managing the data flow from the neural interfaces. Finally, electrical verification of both individual WTBS units and their synchronised operation will be deployed.		

WP Number	3	M4-M44
WP Title	Semantical and Multiphysical Communication for TBS Networking	
Objectives: WP3 aims to develop, test, validate and characterise the performance of a multiphysical communication system between an on-skin interface and the transcalvarial implant. The system will use magnetoelectric, RF near field communications (NFC), ionic sensing as well as optical feedback. The communication system will be managed operationally based on semantic information captured by the optical feedback. WP3 will also provide solutions for networking with multiple implants, network management and network optimization.		
Description of Work and Role of Specific Beneficiaries / Associated partners:		
➤ <i>Task 3.1 - Design and Modelling of Intra-Body and Body-Coupled Communications Systems (Lead: UESSEX, Others: TCD, ALU, TUHH, MINHO, SETU; M4-M28):</i> This task will involve creating a comprehensive model and design of the communication systems that operate within the body (intra-body) and those that link the body to external devices (body-coupled). Will comprise of multiphysical approach to deal with the bio-physical limitations of the body. The design will incorporate four distinct channels, (i) a power channel, (ii) a primary communication channel, (iii) a secondary communication channel and (iv) a feedback channel. The choices for information carriers of each channel are based on (i) magnetoelectric (ii) NFC (iii) ionic (iv) optical. Modelling will incorporate multi physical properties in a integrated manner. ➤ <i>Task 3.2 - Communication networking optimizations (Lead: TCD, Others: UESSEX, ALU, TUHH, MINHO, SETU; M12-M36):</i> We will propose a semantic communication theoretical-based framework, targeting coded information transmission based on stimulation configuration, and compressed sensing to achieve implanted devices' status monitoring. We will exploit the above-mentioned semantic framework to optimise a multi-physical communication system which performs networked power and information transfer, while addressing biophysical human head limitations. We also aim at minimising the volume of data and energy consumption, so that one can increase the number of supportable devices in the in-brain wireless network. ➤ <i>Task 3.3 - Communication Infrastructure Analysis, Simulations, and Validation: multiphysics, semantics, coding, and management (Lead: ALU, Others: UESSEX, TCD, SETU, UKFLR; M24-M48):</i> In this task, the focus will be on assessing the communication framework through a series of rigorous analyses, computer simulations, and real-world validation tests in-vitro and in-vivo. Partners will evaluate the efficiency, resilience, and reliability of the infrastructure, making requisite adjustments and refinements to ensure optimal performance in varied scenarios. We will use CMOS technology integrated with multiphysical transducers. We will work on the detailed design and layout plans for the wireless communication ASIC using CMOS technology for the actual fabrication of the ASIC. We will have a rigorous electrical verification process for both inter-body and external communication systems. Special attention will be given to scenarios involving data collision, ensuring the system can handle such situations effectively and maintain reliable communication tested in-vitro environment (WP5) and demonstrated in-vitro and in-vivo (WP5).		
WP Number	4	M4-M44
WP Title	Preparatory Work for Semantics-based Interference Stimulation	
Objectives: The overall objective is to establish first-in-the-world semantics-based electrical interference stimulation with a network of skull implants. To achieve this, we aim to: (i) improve the state of the art in current flow modeling with novel approaches to model the skull, the skin and the dura compartments, (ii) derive optimal parameters for stimulation using these novel models and an uncertainty analysis with respect to the electrical conductivity and compartmental geometry of the biological tissues, (iii) find optimal sites for implantation in the rodent model of Parkinson's disease, (iv) establish and validate tethered interference stimulation on the rodent PD model (v) simulate the semantics-based interference stimulation.		
Description of Work and Role of Specific Beneficiaries / Associated partners:		
➤ <i>Task 4.1 - Current flow modeling (Lead: TUIL, Others: ALU, MINHO, UKFLR, TUHH, TAU; M4-M28):</i> We will combine high resolution imaging (CT, MRI , in-plane isotropic resolution of 136.7 μm) with the digital twins of the implants and establish novel high-resolution head models of in-vitro neuronal systems, rodents and humans. ➤ <i>Task 4.2 - Uncertainty analysis and optimal parameters (Lead: TUIL, Others: TAU, ALU, MINHO, UKFLR, TUHH; M15-M30):</i> We will model the relevant material and geometrical parameters as distributed random variables using available a priori knowledge and estimate statistical moments by means of a generalised polynomial chaos expansion. We will derive optimal parameters for the implants and provide uncertainty assessment for the digital twins. ➤ <i>Task 4.3 - Strategy for optimal positioning of the implants (Lead: TUIL, Others: TAU, ALU, UKFLR; M24-M48):</i> We will combine the novel current flow models and the results of the uncertainty analysis with a novel optimization approach which takes into account the possibilities of rearrangement of the implants acting as dynamic agents. We will derive optimal implant positions for the proof-of-principle experiments. ➤ <i>Task 4.4 - Design, synthesis and characterization of smart nanoarchitected electrodes by in-vitro validation for interference stimulation (Lead: MINHO, Others: TAU; M6-M36):</i> We will combine the design and optimization of conformal and flexible polymeric-based electrodes for subdural brain implant materials. The electrodes will be produced using several types of polymeric resins and using Additive Manufacturing (AM) techniques like Material Jetting and Vat polymerization. Smart nanoarchitected coating solutions (from elemental to intermetallic transition metals to their nitrides/oxides) and surface plasma functionalization (inert and reactive), including noble metal nanoparticles inclusions will be tested to maximise electrodes base-materials performance and stimulation responses. ➤ <i>Task 4.5 - Establish animal model of PD by 6-OHDA lesioning under ethical considerations (Lead: UKFLR, Others: TUIL; M4-M18):</i> We will extend existing approvals and work closely with full time ethicists and animal caretakers to avoid suffering and maximize patient oriented outcomes. ➤ <i>Task 4.6 - Tethered interference stimulation in animal model of PD (Lead: UKFLR, Others: TUIL; M12-M24):</i> Using the novel implant (WP2), the derived parameters (T4.2), the optimal positions (T4.3), and the established animal model (T4.4), we will perform a proof-of-principle experiment to treat PD in the 6OHDA rat and test it against classical tethered DBS stimulation.		
WP Number	5	M12-M44
WP Title	Integration and Demonstration of Wireless Transcalvarial Brain Stimulators	
Objectives: The overall objective is to build up the first-in-the-world multiscale, multimodal, in-silico, and real life demonstrator environments for a networked transcalvarial brain stimulator (i). Devise in-silico, in-vitro, and preclinical environments to develop, test, and demonstrate the individual components and the integrated Transcalvarial Brain Stimulator (ii). Use the environments in-silico, in-vitro, and in-vivo to develop the project concepts and technologies towards the first in the world novel integrated brain stimulation solution (iii) Demonstrate the technology components; interference stimulation, schematics based stimulation, multiphysics communication, and novel power and data transfer technologies and the integrated system in vitro and in-vivo.		
Description of Work and Role of Specific Beneficiaries / Associated partners:		

- **Task 5.1 - Development of in-silico preclinical platform (Lead: TUIL, Others: TAU, UKFLR, UESSEX; M4-M28):** This task focuses on the development of in-silico computer simulations designed to accurately mimic in-vitro and preclinical scenarios of electric field interference, as well as power and data transfer processes. We will create a detailed virtual model that can predict and analyse the effects and dynamics of electrical fields in controlled experimental settings, aiding in the optimization of these systems before physical testing.
- **Task 5.2 - Development of novel in-vitro 3D neuronal MEA (Lead: TAU, Others: ALU, TUIL, TUHH; M4-M24):** The goal here is to develop an innovative in-vitro 3D neuronal system that is spatially distributed and connectivity controlled. This system will incorporate calcium signalling and multielectrode array measurements, providing a sophisticated platform for testing and demonstrating effectiveness of the stimulator's components under near-physiological conditions.
- **Task 5.3 - Optimize with in-silico simulators the in-vitro and in-vivo experimental environment (Lead: TUIL, Others: TAU, UKFLR; M20-M44):** This task involves the optimization of both in-vitro and in-vivo experimental environments using in-silico simulators. The focus is on refining the stimulus electrode montage and enhancing the interfaces for data and power transfer, to ensure maximum efficiency and effectiveness in the experimental setups.
- **Task 5.4 - In-vitro testing and demonstration of integrated WTBS (Lead: TAU, ALL; M20-M44):** The aim of this task is to conduct in-vitro testing and demonstration of the EFL, wireless communication, and power transfer capabilities within the integrated system. This involves verifying the functionality and efficiency of the wireless interfaces and ensuring that they meet the necessary performance criteria for effective operation.
- **Task 5.5 - In vivo validation of wireless interference stimulation with a rodent model of PD (Lead: UKFLR, Others: ALU, TUIL, TUHH, MINHO; M24-M44):** Two pairs of prototyped Wireless Transcalvarial Brain Stimulators will be deployed in backpacks, carried by PD rats, performing interference stimulation via subcutaneous wires ending in nanostructured and compliant epidural stimulating electrodes. Treatment capabilities of this type of rodent with WTBS "plugs" will be assessed and optimised in Rotarod and Rearing Mobility tests and behaviour assessment.

WP Number	6	M1-M48
WP Title	Networking and Training	

Objectives: (1) Development of an effective Personal Career Development Plan (PCDP) for each Doctoral Candidate (DC). (2) Organisation of training schools, courses, seminars, workshops/special sessions for the DCs (see Section 1.3). (3) Design and implementation of the network-wide training programme to maximise opportunity for learning and cross-fertilization between DCs in different disciplines, institutes and countries through design of the network-wide training events, internal communication infrastructure, and pattern of secondments.

Description of Work and Role of Specific Beneficiaries / Associated partners:

- **Task 6.1 - Organisation and assessment of the DC training program (Lead: TCD, Others: ALL; M1-M48):** Transferable skill training (see Table 1), distribution of educational material to the DCs and outside the network via Web platforms (e.g., a BRAINET dedicated YouTube channel). All partners shall participate in this task, proposing courses/seminars in their specific area of expertise. Development, rollout and implementation of the PCDP for all 14 DCs. Delivery of a module as part of the first training workshop on BRAINET's multi-disciplinary career opportunities, in the context of pre-clinical treatment of neurological disease models by distributed interference stimulation of the brain by a wireless, semantic- and multiphysics-based power and information transfer platform.
- **Task 6.2 - Development of individual training plans and organisation of scientific workshops and special sessions at conferences (Lead: UESSEX, Others: ALL; M1-M48):** Development of individual training plans for each DC and facilitating the provision for training requirements which emerge from the PCDP process. Management of secondments and monitoring of outcomes with associated partner organisations. Organisation and running of scientific workshops and international reputed conferences in the domains of wireless networks and neural interfaces.
- **Task 6.3 - Training and networking execution (Lead: UKFLR, Others: ALL; M1-M48):** The purpose of this task is to oversee the actuation of all the training events, covering courses, seminars and invited talks.

WP Number	7	M1-M48
WP Title	Communication, Dissemination, Exploitation, Regulations and Standards	

Objectives: The main objective of this WP is the dissemination and promotion of the scientific results and achievement of the project and Network. The WP will be responsible for planning and dissemination activities and standardization efforts of the Network.

Description of Work and Role of Specific Beneficiaries / Associated partners:

- **Task 7.1 - Dissemination of Project Results and Achievements (Lead: SETU, Others: ALL; M1-M48):** BRAINET will pay special attention to the dissemination of its achievements at different levels, to ensure that they reach relevant research communities, special interest groups and stakeholders. The BRAINET dissemination plan includes the development of publicity materials (project leaflets among others), website, conference participation, the publication of scientific results in journals, contribution to standardisation, organisation of two workshops and the final project conference (UKFLR), one Industrial Dissemination Day, four activity days and initiatives of interest. The partners will be active in publishing the scientific findings of the project in high impact factor journals and top-ranked conferences. One objective of this task is the preparation of a list of venues to be targeted.
- **Task 7.2 - Exploitation and Standardization Efforts (Lead: UKFLR, Others: ALL; M15-M48):** This task will initially define the list of relevant standardisation bodies and initiatives to target for exploitation. Within this task, participants will closely monitor the progress and discussions of the targeted standards, to keep the project progress aligned and adhering to the standards, and to define possible opportunities for contributions.

3.1.3 Deliverables List

Table 3.1 c Deliverables List

Scientific Deliverables								
No.	Deliverable Title	Short Description	WP No.	Lead	Type	DL	M	
D2.1	Requirement Gathering and Documentation	Living document with increasing granularity describing over arching system requirements.	WP2	UESSEX	R	PU	M3, M6	
D2.2	Biophysical Specifications and Materials Optimization Documentation	Details the development and refinement of microelectronic hardware and material.	WP2	ALU	R	PU	18	
D2.3	Stimulation and Communication Protocols	Includes CMOS-integrable stimulation strategies and inter-WTBS synchronization methods	WP2	SETU	R	PU	30	
D2.4	Design/Verification of Bidirectional Neural Interfaces	Details the design, layout, and electrical verification of CMOS technology interfaces	WP2	UESSEX	R	PU	48	

D2.5	PPy coated nanoporous silicon materials	Materials available for pilot experiments	WP2	TUHH	R	PU	12
D2.6	MECMat materials	PPy coated nanoporous silicon combined with magnetostrictive materials	WP2	TUHH	R	PU	30
D2.7	Prototypes for application	Optimization and scale-up of the materials for applications	WP2	TUHH	R	PU	36
D3.1	Intra-Body and Body-Coupled Communication System Designs and Models	Includes multiphysics and semantic optimization frameworks	WP3	UESSEX	R	PU	24
D3.2	Communication Infrastructure Analysis and Validation	Encompasses simulations, validations, and optimization of network management	WP3	TCD	R	PU	36
D3.3	CMOS-based Wireless Communication Architecture and ASIC Fabrication	Details the design, layout, and electrical verification without and with data collision scenarios	WP3	ALU	R	PU	48
D4.1	Optimized EFI electrode montages in various scales	Details on the EFI electrode, characterization, and functional verification	WP4	TUIL	R	PU	18
D4.2	Electrode optimization and placement	Study of different placement strategies regarding predicted reliable performance	WP4	MINHO	R	PU	36
D4.3	Demonstration of in-silico the human scale EFI and data/power transfer	Study of effectiveness on brain scale of EFI based system and functional verification	WP4	TUIL	R	PU	40
D4.4	Characterization of novel smart functionalized electrodes	Detailed characterization of novel smart functionalized compliant electrodes tailored to response the required frequency range, charge injection capacity, stability and cytotoxicity	WP4	MINHO	R	PU	24
D4.5	Report on 6-OHDA lesion successes by behavioral changes	Details of rat behaviour changes with implantation	WP4	UKFLR	R	PU	12
D4.6	Demonstration on measurable improvement of DBS symptoms by tethered Temporal Interference Stimulation	Details of effectiveness of treatment with stIS	WP4	UKLFR	R	PU	24
D5.1	Development of in-silico optimization of testing and demonstration environments	Integrates in-silico, in-vitro, and in-vivo testing frameworks	WP5	TUIL	R	PU	12
D5.2	In-vitro testing and demonstration 3d neuronal system with distributed connectivity	Includes in-vitro testing and system capabilities demonstration	WP5	TAU	R	PU	24
D5.3	Demonstration of the integrated system in-vitro	Assesses treatment capabilities and system performance in a rodent model of PD	WP5	TAU	R	PU	32
D5.4	Demonstration of the integrated system in-vivo	Assesses treatment capabilities and system performance in a rodent model of PD	WP5	UKFLR	R	PU	36
Management, Training and Dissemination Deliverables							
No.	Deliverable Title	Short Description	WP No.	Lead	Type	DL	M
D1.1	Annual Reports on General Management	Report on Geneneral Management	WP1	UKLFR	R	CO	M12, M24, M36, M48
D1.2	Technical Reports on Cross-WP Coordinated Work	Reports on work related to multiple WPs	WP1	UKLFR	R	CO	M18, M30, M40
D1.3	Risk assessment and contingency planning, regular re-assessment	Report of general risk of BRAINET	WP1	UKLFR	R	CO	M3, M15, M30
D1.4	Valorization Report	Report on valorization of BRAINET	WP1	UKLFR	R	CO	M24, M48
D1.5	Data Management Planning	Report on the data management plan of BRAINET	WP1	UKLFR	R	CO	M6, M30
D6.1	Report on Network-Wide Training Program and Personal Career Development Plans	Includes transferable skill training and educational material distribution	WP6	TCD	R	CO	M12
D6.2	Scientific Workshops and Special Sessions Compilation	Details the organization, content, and outcomes of training events	WP6	UESSEX	R	CO	M24
D6.3	Report on Secondment Program and Cross-Disciplinary Learning Outcomes	Documents the experiences and skill acquisition of Doctoral Candidates	WP6	TCD	R	CO	M36
D7.1	Communication and Dissemination Strategy	Includes public engagement activities, development of publicity materials, and conference participation	WP7	ALU	R	CO	12
D7.2	Exploitation Plan and Standardization Efforts	Focuses on the inclusion of scientific findings in international standards and contributions to relevant standardization bodies	WP7	TUHH	R	CO	24
D7.3	IP Management and Regulatory Compliance Guide	Details knowledge management, regulatory adherence, and networking activities	WP7	TUIL	R	CO	36

3.1.4 Milestones List

Table 3.1 d Milestones List

No.	Title	WP	Lead	D	Means of Verification
M1.1	Establishment of Project Management Framework	WP1	UKLFR	M12	Project management documents and minutes from the consortium-wide meeting
M1.2	Mid-Project Management Review	WP1	UKLFR	M24	Review report and updated project management plan

No.	Title	WP	Lead	D	Means of Verification
M2.1	Completion of Microelectronic Design and Simulation; Fab-transfer for fabrication	WP2	ALU	M16	EDA verification and application notes
M2.2	Completion of Microelectronic Hardware Prototype	WP2	ALU	M24	Prototype demonstration and datasheet with technical documentation
M2.3	Integration of WTBS Devices	WP2	ALU	M36	Integration report and system validation results
M3.1	Validation of Intra-Body Communication Models	WP3	UESSEX	M24	Simulation results and validation report
M3.2	Optimized Communication Infrastructure Implementation	WP3	UESSEX	M36	Deployment documentation and performance analysis report
M4.1	Development of Current Flow Models	WP4	TUIL	M24	Model specifications and parameter determination documentation
M4.2	In-Vivo Validation of Interference Stimulation	WP4	TUIL	M36	Validation study results and experimental data
M5.1	Integration of Multiscale Demonstration Environments	WP5	TAU	M24	Integration report and demonstration environment specs
M5.2	In-Vivo Demonstration of Integrated System	WP5	TAU	M48	Demonstration results and in-vivo testing documentation
M6.1	Launch of Network-Wide Training Program	WP6	TCD	M12	Training program agenda and PCDP documents
M6.2	Mid-Term Training Impact Assessment	WP6	TCD	M36	Assessment report and feedback from participants
M7.1	Initial Dissemination and Communication Strategy Release	WP7	SETU	M12	Strategy document and communication materials
M7.2	Draft of Exploitation Plan and Standardization Engagement	WP7	SETU	M24	Draft exploitation plan and records of engagement with standardization bodies

3.1.5 Recruitment Table per Beneficiary

Table 3.1 e Recruitment Table per Beneficiary

Researcher No.	Recruiting Participant (short name)	PhD awarding entities	Planned Month 0-45	Start	Duration (months) 3-36
1	ALU	ALU	M6		36
2	ALU	ALU	M6		36
3	MINHO	MINHO	M6		36
4	UESSEX	UESSEX	M6		36
5	UESSEX	UESSEX	M6		36
6	SETU	SETU	M3		36
7	TCD	TCD	M6		36
8	TCD	TCD	M6		36
9	TUIL	TUIL	M3		36
10	TAU	TAU	M3		36
11	TUHH	TUHH	M3		36
12	MINHO	MINHO	M6		36
13	UKLFR	ALU	M6		36
14	TAU	TAU	M3		36

3.1.6 Individual Research Projects, including secondment plan

Table 3.1 f Individual Research Projects

Fellow	Host	PhD enrolment	Start	Duration	Deliverables
DC1	ALU	Yes	M6	36	D3.1, D3.2, D3.3, D5.2, D5.3, D6.1, D6.2 and D6.3
Development and fabrication of wireless multiphysical communication ASICs					
Objectives : (i) Development of Communication Concept and Architecture for WTBS Network, Development and Design of Communication Hardware for WTBS and Fabrication and Testing of ASICs for WTBS.					
Expected Results : (i) Communication concepts, architectures, and protocols for semantic communication between WTBS implants and external units (ii) CMOS-based wireless communication architecture with multiphysical transmission concepts for at least two independent transmission channels. (iii) Semantic wireless communication ASIC suitable for integration into the WTBS device.					
Planned secondment(s) : TU Delft, Prof. Dante Muratore, during year 1, 3-6 months to gather insights and hands-on experience in compressive high-bandwidth transmission of neural data in in-vitro and in-vivo settings.					
Enrolment in Doctoral degree(s) : DC1 will be enrolled in the University of Freiburg, Germany, for the Ph.D. degree program. Dual supervision will be considered for the DC1 Supervision.					
DC2	ALU	Yes	M6	36	D3.1, D3.2, D3.3, D5.2, D5.3, D6.1, D6.2 and D6.3
Development and fabrication of bidirectional neural interface ASICs with arbitrary waveform stimulators					
Objectives : (i) Communication concepts, architectures, and protocols for inter-WTBS communication and synchronization (ii) Bidirectional neural interface architecture suitable for independent and connected stimulation and recordings. (iii) Conceptual architecture of a neural data pre-processor to be integrated into each WTBS (iv) Neural stimulation and recording ASIC, suitable for integration into the WTBS device.					
Expected Results : (i) Communication concepts, architectures, and protocols for semantic communication between WTBS implants and external units (ii) CMOS-based wireless communication architecture with multiphysical transmission concepts for at least two independent transmission channels. (iii) Semantic wireless communication ASIC suitable for integration into the WTBS device.					
Planned secondment(s) : MED-EL (Markus Oberpaleiter) , during year 1, 3-6 months to learn about multichannel human neural interfaces and be informed about security measures and regulations of implant systems.					
Enrolment in Doctoral degree(s) : DC2 will be enrolled in the University of Freiburg, Germany, for the Ph.D. degree program. Dual supervision will be considered for the DC2 Supervision.					
DC3	MINI	Yes	M6	36	D2.4, D2.6, D 4.4, D6.1, D6.2 and D6.3
Development of subdural brain implants by additive manufacturing					
Objectives : Design and development of a polymeric micro part to support the nanoarchitected coating solutions and other surface functionalization approaches for subdural brain implant materials; Developed new polymeric resin for vat polymerization with focus in brain implant application and Evaluation of the integrated solution: material and design polymeric part.					

Expected Results : (a) Polymeric micro part to be implemented in subdural brain implants. (b) Selection of best polymeric resin for Additive Manufacturing according to the specificity of interaction of materials with the brain. (c) Comprehensive assessment of the polymeric part and electrode system.					
Planned secondment(s) : Prof Jens Hauelsen's laboratory, University of Ilmenau, Germany during 3 months for signal acquisition and processing studies; Prof Ulrich Hofmann laboratory, University of Freiburg, Germany during 3 months for in-vivo preclinical testing.					
Enrolment in Doctoral degree(s) : DC3 will be enrolled in the University of Minho, Portugal, for the Ph.D. degree program. Dual supervision will be considered for the DC3 Supervision.					
DC4	UESSEX	Yes	M6	36	D2.1, D2.3, D3.1, D3.2, D6.1, D6.2 and D6.3
Multiphysical communications using magnetoelectric, NFC, ionic media and optics for networked power and information transfer to overcome biophysical human head limitations					
Objectives : Investigate the Feasibility of Multiphysical Communications, Characterise the biophysical limitations of Human Heads and Develop an Optimised Multiphysical Communication System.					
Expected Results : (a) Demonstrate the feasibility of magnetoelectric, NFC, ionic media and optics communications for networked power and information transfer for D3.1; (b) Obtain comprehensive understanding of the biophysical limitations of human heads concerning magnetoelectric, NFC, ionic media and optics communications for D2.1; (c) Create an optimised magnetoelectric, NFC, ionic media and optics communication system tailored for networked power and information transfer, considering the biophysical human head limitations for D3.1, D2.3 and D3.2. Results will be published in two conference papers, and three top-ranked journal papers.					
Planned secondment(s) : DC 4 will spend 3 months in INRIA at M18, France, hosted by Dr Malcolm Egan, to integrate robust models of information and power transfer into distributed multiphysical channels. And 3 months in Aalborg University, Denmark, at M12, for understanding pre-clinical requirements of wireless multiphysics for implantable devices.					
Enrolment in Doctoral degree(s) : DC4 will be enrolled in the University of Essex, UK, for the Ph.D. degree program. Dual supervision will be considered for the DC4 Supervision.					
DC5	UESSEX	Yes	M6	36	D2.3, D3.1, D3.2, D3.3, D6.1, D6.2 and D6.3
Multimodal communication system for subdural brain implants: enabling networking					
Objectives : Develop a Robust Multimodal Communication Framework, Optimise Data Transmission and Bandwidth Utilisation and Evaluate Safety, Reliability, and Long-Term Viability.					
Expected Results : (a) A functional multimodal communication framework made just for subdural brain implants that can use neural signals and other physiological signals to send and receive data reliably for D3.1, D3.3 and D2.3. (b) Efficient signal processing and data compression algorithms D3.2. Results will be published in two conference papers, and three top-ranked journal papers.					
Planned secondment(s) : DC 5 will spend 3 months in INRIA at M12, France, hosted by Dr Malcolm Egan, to model neuronal data compression algorithms for coded spiking pattern transmission and 3 months at M30 at G.TEC, for measuring feasibility of developed signal process for existing implantable devices.					
Enrolment in Doctoral degree(s) : DC5 will be enrolled in the University of Essex, UK, for the Ph.D. degree program. Dual supervision will be considered for the DC5 Supervision.					
DC6	SETU	Yes	M3	36	D3.2, D3.3, D6.1, D6.2 and D6.3
Self-Optimizing and Self-Healing Implanted Communication Network					
Objectives : Modelling of each Transcalvarial Networked Brain Stimulator (WTBS), Self-Optimization of the network of WTBS and Failure-resilience and self-healing of implanted networks.					
Expected Results : (i) Theoretical formulation and modelling of the implant communication network (ii) Design of the implant network and autonomous resource optimization for the network endpoints and links (iii) Autonomous and efficient failure-resilience and healing of the implant network.					
Planned secondment(s) : DC 6 will spend 3 months in Purposeful at M20, Greece, hosted by Dr Georgia Tsiliki, to model neuronal data compression algorithms for coded spiking pattern transmission and 3 months at M12 at ALU, Prof Matthias Kuhl, for hardware requirements for self-optimisation and self-healing mechanisms.					
Enrolment in Doctoral degree(s) : DC6 will be enrolled in the South East Technological University, Ireland, for the Ph.D. degree program. Dual supervision will be considered for the DC6 Supervision.					
DC7	TCD	Yes	M6	36	D3.1, D3.2, D6.1, D6.2 and D6.3
Semantics-based simultaneous wireless power and information transfer technologies					
Objectives : Design a semantics-based framework, aiming to sample states of brain functions/regions and transfer data and power based on it, Define semantics-based communication and wireless powering strategies and Investigate suitable network architectures for semantics-based information and power transfer.					
Expected Results : (a) A semantic communication-based model for the interplay between wireless power and information transfer tasks for distributed interference stimulation, for D3.1; (b) Reduce the volume of data and energy consumption, increasing the number of supportable devices in the in-brain wireless network, for D3.2; (c) Capture in a more meaningful and efficient way the goals of in-brain control (actuation/stimulation) and sensing (monitoring/recording), for D3.1, D3.2. Results will be published in three conference papers, and three top-ranked journal papers.					
Planned secondment(s) : DC 7 will spend 3 months in TAU at M12, Finland, hosted by Prof Jari Hyttinen, to investigate how practical constraints related to distributed interference stimulation can be incorporated in the semantic communication model. DC 7 will also spend 3 months in G.TEC at M24, Austria, hosted by Dr Christoph Guger, to optimise how the proposed model takes into account stimulation and monitoring functionalities in a BMI context.					
Enrolment in Doctoral degree(s) : DC7 will be enrolled in the Trinity College Dublin, Ireland, for the Ph.D. degree program. Dual supervision will be considered for the DC7 Supervision.					
DC8	TCD	Yes	M6	36	D3.1, D3.2, D3.3, D6.1, D6.2 and D6.3
Development and fabrication of bidirectional neural interface ASICs with arbitrary waveform stimulators					
Objectives : (a) The external transceiver will transmit to the implants coded information based on the stimulation configuration, based on user-specified parameters, via the use of a secondary wireless magnetic channel; (b) We will adopt a compressed sensing approach to achieve implanted devices' status monitoring, via the use of a secondary wireless magnetic channel. (c) The downlink provided power and time-delay, phase- or frequency-coded stimulation commands and the corresponding uplink monitored stimulation state information, will enable a multi-scale network-level control approach.					
Expected Results : (a) Coded information transmission based on stimulation configuration, via the use of a secondary wireless magnetic channel, for D3.1; (b) Compressed sensing approach to achieve implanted devices' status monitoring, via the use of a secondary wireless magnetic channel, for D3.1, D3.2; (c) Multi-scale network-level control approach, to monitor how different implants affect each other, for D3.2, D3.3. Results will be published in three conference papers, and three top-ranked journal papers.					
Planned secondment(s) : DC8 will spend 3 months in UESSEX at M12, UK, hosted by Prof Michael Barros, to investigate which signal processing algorithms to apply compressed sensing to monitoring. DC 8 will also spend 3 months in INRIA at M24, France, hosted by Dr Malcolm Egan, to study how to efficiently manage the tradeoff between cost minimization and identification in multi-scale network-level control.					

Enrolment in Doctoral degree(s) : DC8 will be enrolled in the Trinity College Dublin, Ireland, for the Ph.D. degree program. Dual supervision will be considered for the DC8 Supervision.					
DC9	TUIL	Yes	M6	36	D4.1, D4.2, D4.3, D5.1, D6.1, D6.2 and D6.3
Current flow modelling in the head					
Objectives : Development of Advanced Current Flow Models, Comprehensive Uncertainty Analysis in Electrical Conductivity and Tissue Geometry and Identification of Optimal Implantation Sites.					
Expected Results : (i) Digital twin of the implant and current flow models of the individual head with multi-layer skull models (ii) Uncertainty analysis for the novel current flow models identifying major coefficients (iii) Optimization strategy for positioning the implants and distributing the currents based on target regions in the brain.					
Planned secondment(s) : TAU, Prof. Hyttinen, M15-17, 3 months, development of in-silico simulation setup; MINHO, Prof. Vaz, M25-M27, 3 months, in-silico modelling of electrodes and interfaces.					
Enrolment in Doctoral degree(s) : DC9 will be enrolled in the Technical University Ilmenau, Germany, for the Ph.D. degree program. Dual supervision will be considered for the DC9 Supervision.					
DC10	TAU	Yes	M3	36	D4.2, D4.3, D4.6, D5.2, D5.3, D6.1, D6.2 and D6.3
Develop methods to assess the neuronal system (brain) state and the effect of the spatiotemporally steered stimulation on it for the control strategies of the spatiotemporal interference stimulation					
Objectives : Develop measures and metrics for the neuronal system state assessment, Assess and demonstrate the tools using the in-vitro 3D neuronal model platform with spatiotemporal stimulation and Integrate the tools with the spatiotemporal networked stimulation and communication technology and demonstrate the use in preclinical animal model.					
Expected Results : (i) Theoretical and practical spatiotemporal brain interaction assessment methods (ii) Set of tools to assess the spatiotemporal brain state, stimulation need and stimulation outcomes (iii) Novel insight into spatio temporal brain stimulation control and use in PD and beyond.					
Planned secondment(s) : Prof Ulrich Hofman laboratory, University of Freiburg, Germany for in-vivo preclinical studies for 3 months, Prof Barros lab to integrate the neuronal status assessment with multi-physical communication platform for another 3 months.					
Enrolment in Doctoral degree(s) : DC10 will be enrolled in the Tampere University, Finland, for the Ph.D. degree program. Dual supervision will be considered for the DC10 Supervision.					
DC11	MINHO	Yes	M6	36	D4.4, D6.1, D6.2 and D6.3
Functionalization of electrode materials for subdural brain implants					
Objectives : Develop smart coating architectures and a surface plasma patterning, Optimise the response of functionalized subdural brain implant materials, Evaluate Safety, Reliability, and Long-Term Response of Functionalized Electrodes.					
Expected Results : (a) Nano-functionalized surfaces of electrodes that will be designed and implemented in subdural brain implants. (b) Selection of best surface plasma texturing protocols and thin film system according to the specificity of a barely explored interaction of materials with the brain. (c) Comprehensive safety assessment of the electrode system.					
Planned secondment(s) : Prof Jens Hauelsen's laboratory, University of Ilmenau, Germany during 3 months for signal acquisition and processing studies; Prof Ulrich Hofmann laboratory, University of Freiburg, Germany during 3 months for in-vivo preclinical testing.					
Enrolment in Doctoral degree(s) : DC11 will be enrolled in the University of Minho, Portugal, for the Ph.D. degree program. Dual supervision will be considered for the DC11 Supervision.					
DC12	TUHH	Yes	M6	36	D2.4, D2.5, D2.6,, D6.1, D6.2 and D6.3, D7.1
Development & fabrication of hybrid materials with integrated Magnetoelectric energy conversion					
Objectives : Developing the material concept based on nanoporous silicon hybrid materials, Combining the PPY-nanoporous hybrid with a magneto-to-electrical energy materials and Fabrication and testing of MECMats.					
Expected Results : (i) Demonstration of material integrated mechanical-to-electrical energy conversion and conversion power optimization (ii) Combining mechanical-to-electrical with magnetic-to-mechanical energy conversion and corresponding hybrid materials optimization. (iii) Production of MECMats for integration as wireless energy supply materials for BRAINET devices.					
Planned secondment(s) : ALU, Prof. Kuhl, M12, 3 months, modelling of electrodes and interfaces, integration into hardware and MultiChannel Systems, M20, 3 months, for pre-clinical requirements of wireless body radiation.					
Enrolment in Doctoral degree(s) : DC12 will be enrolled in the Hamburg University of Technology, Germany, for the Ph.D. degree program. Dual supervision will be considered for the DC12 Supervision.					
DC13	UKLFR	Yes	M3	36	D4.5, D4.6, D5.5, D6.1, D6.2 and D6.3
Pre-clinical Validation of Temporal Interference Stimulation with a Parkinsonian Rat Model					
Objectives : The DC will learn to use a robust hemi-parkinsonian rat model, assess intracortical electrophysiological changes based on tethered interference stimulation, validate PD symptom improvement of invasive DBS and low-invasive tethered interference stimulation, assess histological consequences of localised TIS and validate wireless interference stimulation nodes in PD animals. Special consideration towards potential clinical use of TIS (material effects, frequencies, strength, orientation, placement) will be given.					
Expected Results : (a) Establish a robust animal model of Parkinson's Disease (D4.5); (b) Get detailed insight into therapeutic temporal interference stimulation in vivo D4.6 (c) Demonstrate tethered and wireless interference stimulation to treat PD in an animal model (D5.5).					
Planned secondment(s) : TAU, Jari Hyttinen, M9, 3 months, familiarise with in vitro TIS setup MED-EL, Werner Lindenthaler, M24, 3 months, learn about documentation and quality control issues in electrical stimulation industry.					
Enrolment in Doctoral degree(s) : DC13 will be enrolled in the Technical Faculty of the University of Freiburg, Germany, for the Dr.rer.nat or the Dr.-Ing. degree program. Biologists by training may enrol into the Dr. hum. biol. the program of the Medical Faculty. Dual supervision will be considered for the DC13 Supervision.					
DC14	TAU	Yes	M3	36	D4.6, D5.1, D5.2, D5.3, D6.1, D6.2 and D6.3
Development and demonstration of 3D-in-vitro environment for of electric field interference neuronal stimulation with self-organized networked electrode array					
Objectives : Development and demonstration of 3D in-vitro biological neuronal test environment with multielectrode electric interference stimulation, Optimize the capabilities of multielectrode electric interference stimulation (spatial and temporal properties) in-vitro and Integration and demonstration of the multimodal communication and self-organized sensory network in-vitro.					
Expected Results : (i) 3D neuronal spatially distributed in-vitro model for multielectrode electric field interference measurements; (ii) Theoretical and practical temporal and spatial characteristic of multielectrode based electric field interference stimulation scheme; (iii) Demonstration of the multimodal neuronal communication system in-vitro with insight towards human scale and with human 3D slices at university of Freiburg.					
Planned secondment(s) : Prof Ulrich Hofman laboratory at M22, for 3 months, University of Freiburg, Germany to test the in-vitro environmen twitch human ex-vivo 3D slice cultures, and NEUROCONN, Germany, at M25, for 3 months, in pre-clinical models for in-vitro testing of implants.					
Enrolment in Doctoral degree(s) : DC14 will be enrolled in the Tampere University, Finland, for the Ph.D. degree program. Dual supervision will be considered for the DC14 Supervision.					

3.1.7 Progress monitoring and evaluation of individual research projects

In order to assess the success of the projects, qualitative and quantitative indicators are required. Participation in conferences, a number of publications, a number of patents, and a number of new collaborations are all quantitative indicators used to evaluate research activities. *Quantitative indicators* for assessing knowledge transfer include the following: i) the rate of recruited researchers, ii) the number and involvement of researchers, iii) the number of training events, iv) the number of master or doctoral theses supervised, and v) the number of DCs attending each training course. *Qualitative indicators* for assessing knowledge transfer include i) the level of satisfaction of recruited DCs, ii) monitoring the original knowledge transfer plan, and iii) the quality of inter-sectoral collaborations. *HR Excellence* - Finally, the vast majority of research and academic institutions involved in the project have been positively evaluated for obtaining the logo for HR excellence in research, implying that the institutions have already put in place mechanisms to monitor, evaluate, mentor, and coach the PhD students enrolled in their training programmes.

3.1.8 Implementation Risks#@RSK-MGT-RM@#

BRAINET, a visionary research project, acknowledges key risks in implementation, addressed through WP1 monitoring. The coordinator ensures timely solutions, focusing on joint projects, technical challenges, partner issues, expertise, agreements, dissemination, market/user, and competition risks. A comprehensive risk assessment in M1-3, with ongoing mitigation by the PC and SB, ensures project success.

Table 3.1 g Implementation Risks

Description of risk (indicate level of (i) likelihood, and (ii) severity: Low/Medium/High)	WP	Proposed risk-mitigation measures
Scientific Risks		
Nanomodification of known materials is insufficient for Temporal Interference Stimulation (medium/low)	WP2,4	Move to bulk conducting, 3D additive manufactured electrode systems
Multiphysics offer too much complexity for the ASIC chip design and impact its reliable operations (Medium/High)	WP2,3	Decrease the amount of channels based on the tolerated ASIC multiple channel processing capabilities. Integrate secondary channels functionalities into the primary channel.
Microelectronics of TWBS hub has too high an energy consumption to be driven by ME power stack (medium/high)	WP 2	Change CMOS process and increase and increase number of power stack for parallel use
Power stack materials provide insufficient power (medium/low)	WP 2	Move to multiple, parallel stack, change compositions by using less innovative, published materials
Semantic coding doesn't provide the expected energy saving capacities (low/medium)	WP 3	Move to standard Shannon type coding or compressed sensing approaches
Temporal Interference Stimulation doesn't produce symptom relief in PD animal model (medium/low)	WP 5	Implement frequency hopping or multi frequency TIS
Training Risks		
Problem in training and seminar (Low,Low)	WP6	If some partners are unable to organize planned training programs or seminars, alternatives will be provided in a very short time frame.
Management Risks		
Delays in some tasks cause delays in depending on tasks (Medium/Medium)	ALL	The Project Manager has the responsibility to continuously monitor the status of the project through reports and meetings. In case of impacts on other project tasks (in terms of time delays), alternative solutions and a re-planning of the affected tasks must be initiated immediately.
A partner leaves the consortium in the course of the project, so crucial knowledge crucial to the project's success is missing. (Low/High)	ALL	This project is of central importance for each partner, partners have some overlap expertise, so a partner defecting could be substituted.
Departure of DCs (Low, Medium)	ALL	The positions will also be advertised again (with remaining work done by new DC) in sufficient media, including newspapers, mailing lists, and job websites. Thus, a well-defined advertisement process is included in the program.

#\$RSK-MGT-RM\$#

3.1.9 Recruitment strategy

BRAINET partners, committed to equal opportunities, adhere to the European Charter for Researchers and Code of Conduct. Quarterly project meetings include gender balance reviews, ensuring active implementation and monitoring of gender-related activities in decision-making & secondment planning throughout the project.

◆ *A Recruitment Committee (RC)* comprising the DC supervisor and 6 members, ensures expertise, gender balance, and adherence to recruitment laws. UK PhD student recruitment follows public calls and professorial committees. The RC, within the BRAINET Network, ensures at least 50% female DCs for gender equality. BRAINET prioritizes gender and minority equality in hiring. The final list of DCs aligns with project requirements by M1, approved by the SB in M2. Advertisements, reflecting job details and European Charter compliance, run for a minimum of two months. Applicants submit CVs and cover letters, emphasizing transparency and inclusivity in the recruitment process.

◆ **Hiring and Selection:** Applicants meeting minimum requirements undergo initial shortlisting, with rejections communicated by email. The RC, comprised of experts, evaluates eligible candidates based on grades, publications, awards, and research potential. RC members provide individual assessments, and a collective mark scheme determines overall rankings. Mobility experiences and previous research/internships enhance candidates' profiles. The RC, holding conference calls if necessary, finalizes a ranked list for the Steering Board (SB) to approve. Gender equality is considered in the ranking. The SB's decision leads to contract signing by selected candidates, aligning with National Legislation requiring shared facilities for recruited DCs.

3.2 Quality, capacity and role of each participant

3.2.1 Appropriateness of the infrastructure and capacity of each participating organisation

All Network partners are established organizations equipped with necessary facilities for the proposed training program. Academic and industrial partners have essential departments, ensuring smooth project execution. Prior engagement in multi-partner projects reflects their experience and capabilities. Legal and financial departments are well-organized and experienced. Compliance with the European Charter for Researchers guarantees independent, adequately equipped premises for hosting DCs. As the coordinating partner, UKFLR, with extensive involvement in EC-funded projects, is adept at coordination and resolving issues related to EC rules or the training network. Full details are available in Sections 4 and 5 of the proposal.

3.2.2 Consortium composition and exploitation of partners' complementarities

BRAINET recognizes the critical role of collaborative efforts across diverse research domains, encompassing in-silico, in-vitro, and in-vivo neuroengineering methods, for the success of the proposed project. The consortium composition, carefully chosen, adheres to the following principles:

- ◆ Formation of a collaborative consortium comprising industrial leaders and esteemed academic institutions, ensuring excellence in neuroengineering, communications, and smart materials expertise.
- ◆ Establishment of a geographically diverse consortium to provide a unique and enriching experience for DCs from various countries, fostering diversity in cultures, work ethics, and environments.
- ◆ Commitment to addressing a broad spectrum of research fields and technical challenges integral to advancing neuroengineering for next-generation wireless brain interfaces.

Industry participants will contribute practical knowledge, real-life applications, clinical-driven insights, and prototypes for experimentation. Academic partners will provide expertise, customized tools, and simulation resources for proof-of-concept. Noteworthy industrial partners, such as MED-EL specializing in neuroprosthetics like the Cochlea Implant, bring market insights and practical experimentation, offering transferable skills crucial for DCs' successful careers in neuroengineering. Each partner's expertise related to their involvement in Work Packages is detailed in the accompanying table. Validation of partners' expertise is supported by the publications outlined in Section 4.

Table 3.2a AP Contribution

Partner	Field of research training
NC	High quality, low noise electronics for actual clinical use. Certification, documentation and quality control according to MDR. Distributed recording and big data analysis
GTEC	Innovative EEG oriented devices for clinical use and therapy. Certification, documentation and quality control according to MDR. Distributed recording and closed loop data analysis of non-invasive recordings
MCS	Innovative intracortical oriented devices for research and neuroscience. Certification, documentation and quality control according to GLP. Big data handling and analysis
CORT	Innovative peripheral nerve oriented devices for research and neuroscience. Certification, documentation and quality control according to GLP for later clinical use. Materials science and production technology
MED-EL	Wide range of stimulation oriented research in the central and peripheral nervous system. Certification, documentation and quality control according to GLP for later clinical use. Materials science and production technology.
PIKE	Innovative repurposing of pharmaceuticals, certification, documentation and quality control according to IVD-MDR. Real world economic planning and supply chain management

3.2.3 Commitment of beneficiaries and associated partners to the programme

All partners, including associate partners, are fully dedicated to the project, participating in hosting, secondments, and all research and training activities. Beneficiary partners, deeply engaged and committed, will host 14 DCs for six-month secondments. Partner involvement is evident in organizing training events, courses, and seminars, ensuring a high level of commitment. Principal supervisors will dedicate at least 15% of their time to DC supervision. The project partners, experienced in MSCA projects, demonstrate full commitment, as evidenced by their successful completion of numerous projects.

##\$QUA-LIT-QL\$## ##\$WRK-PLA-WP\$##